# GONALITY AND QUINTIC POINTS ON INTERMEDIATE MODULAR CURVES

MARIN VARIVODA

Abstract. For $\{\pm 1\} \subseteq \Delta \subseteq (\mathbb{Z}/N\mathbb{Z})^\times$ let $X_\Delta(N)$ be the intermediate modular curve between $X_1(N)$ and $X_0(N)$. We determine the $\mathbb{Q}$-gonality of every strict intermediate curve $X_\Delta(N)$ with $N \leq 50$, and of $X_1(N)$ for $41 \leq N \leq 50$, where it equals the upper bound of Derickx and van Hoeij at every level. We also settle the thirty levels left open by Derickx, Hwang, Jeon and Orlić, completing the list of curves $X_0(N)$ with infinitely many points of degree 5, and we show that exactly 43 strict intermediate curves have infinitely many points of degree 5. As a consequence, the strict intermediate curves of minimum density degree 5 are exactly the four $\mathbb{Q}$-pentagonal curves found by Orlić. The proofs use published classifications, the Castelnuovo–Severi inequality, and three computational methods; all computations can be rerun from the accompanying code.

## 1. Introduction

Let $N$ be a positive integer and $\Delta$ a subgroup with $\{\pm 1\} \subseteq \Delta \subseteq (\mathbb{Z}/N\mathbb{Z})^\times$. The intermediate modular curve $X_\Delta(N)$ is the modular curve attached to

$$\Gamma_\Delta(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : c \equiv 0 \bmod N,\ a \bmod N \in \Delta \right\}.$$

It lies between $X_1(N) = X_{\{\pm 1\}}(N)$ and $X_0(N) = X_{(\mathbb{Z}/N\mathbb{Z})^\times}(N)$, and it is defined over $\mathbb{Q}$. We call $X_\Delta(N)$ *strict* if both inclusions $\{\pm 1\} \subsetneq \Delta \subsetneq (\mathbb{Z}/N\mathbb{Z})^\times$ are strict. We follow the notation of Jeon, Kim and Schweizer [25], Orlić [37] and Derickx and Orlić [13], and we name curves by their LMFDB labels [42, 39].

For a curve $C$ over a field $k$, the *$k$-gonality* $\mathrm{gon}_k C$ is the least degree of a nonconstant morphism $C \to \mathbb{P}^1$ defined over $k$. A *point of degree $d$* on $C/\mathbb{Q}$ is a closed point whose residue field has degree $d$ over $\mathbb{Q}$. The *density degree set* $\delta(C/\mathbb{Q})$ is the set of $d$ such that $C$ has infinitely many points of degree $d$, and its least element $\min \delta(C/\mathbb{Q})$ is the *minimum density degree* of $C$ [26]. By Hilbert irreducibility, $C$ has infinitely many points of degree $\mathrm{gon}_{\mathbb{Q}} C$, so $\min \delta(C/\mathbb{Q}) \leq \mathrm{gon}_{\mathbb{Q}} C$.

1.1. **Previous work.** Points of low degree on modular curves have been studied intensively, with the degree going up one at a time. For $X_0(N)$, the curves with infinitely many points of degree 2 follow from Ogg's classification of hyperelliptic curves [36] and Bars' classification of bielliptic curves [4]; degree 3 is due to Jeon [23]; degree 4 to Hwang and Jeon [21] and independently to Derickx and Orlić [12]; and degree 5 to Derickx, Hwang, Jeon and Orlić [11], except for thirty levels that their methods did not reach. For intermediate curves, degree 2 is due to Jeon, Kim and Schweizer [25], degree 3 to Dalal [6], and degree 4 to Derickx and Orlić [13]. In [13] the authors explain the obstacles to extending their methods to degree 5; Theorem B below settles that case.

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For gonality, the hyperelliptic and trigonal intermediate curves were found by Ishii and Momose [22] and Jeon and Kim [24]; the list of Ishii and Momose misses the hyperelliptic curve $X_\Delta(21)$ with $\Delta = \langle -1, 8\rangle$, see [24, 25]. Orlić [37] determined all intermediate curves of $\mathbb{Q}$-gonality at most 5, and the $\mathbb{Q}$-gonality of every strict curve with $N \le 40$. The $\mathbb{Q}$-gonality of $X_0(N)$ is known for $N < 145$ by Najman and Orlić [34], and that of $X_1(N)$ for $N \le 40$ by Derickx and van Hoeij [16], who also gave upper bounds for $N \le 250$.

1.2. **Results.**

**Theorem C.** *Let $X_\Delta(N)$ be a strict intermediate curve with $N \le 50$. Its $\mathbb{Q}$-gonality is given in Table 3. For $N \le 40$ these values are due to Orlić* [37]*, as are those of* 6 *of the* 24 *curves with $41 \le N \le 50$, which have gonality* 4 *or* 5*; two more have genus* 3 *and* 4 *and are trigonal. The values for the other* 16 *curves are new.*

By Orlić's classification of curves of $\mathbb{Q}$-gonality at most 5 [37], these 16 curves have $\mathbb{Q}$-gonality at least 6; for the five of them with $\mathbb{Q}$-gonality 6, the new part is the upper bound.

**Theorem D.** *For $41 \le N \le 50$, the $\mathbb{Q}$-gonality of $X_1(N)$ equals the upper bound of Derickx and van Hoeij* [16]*:*

| $N$ | *41* | *42* | *43* | *44* | *45* | *46* | *47* | *48* | *49* | *50* |
|---|---|---|---|---|---|---|---|---|---|---|
| $\mathrm{gon}_{\mathbb{Q}} X_1(N)$ | *22* | *12* | *24* | *15* | *18* | *19* | *29* | *16* | *21* | *15* |

Previously, Derickx and van Hoeij [16, §5] showed that $\mathrm{gon}_{\mathbb{Q}} X_1(N) > 8$ for $41 \le N \le 50$ and that $\mathrm{gon}_{\mathbb{Q}} X_1(43) \ge 17$, and Najman and the author [35, Proposition 5.4 and Table 1] obtained the lower bounds $\mathrm{gon}_{\mathbb{F}_p} X_1(N) \ge 10$ over finite fields for $N \in \{41, 42, 44, 45, 46, 48, 50\}$, and $\ge 19$ for $N = 43$.

**Theorem A.** *For each of the thirty levels*

$$N \in \{74, 77, 79, 82, 83, 85, 86, 88, 89, 92, 96, 98, 99, 100, 101,$$
$$103, 107, 111, 118, 123, 128, 131, 141, 142, 143, 145, 155, 159, 167, 191\}$$

*left open in* [11, Remark 1.11 and §7]*, the curve $X_0(N)$ has only finitely many points of degree* 5*, and no morphism of degree* 5 *to $\mathbb{P}^1$ over $\mathbb{Q}$. Together with* [11, Theorem 1.10]*, this completes the list of curves $X_0(N)$ with infinitely many points of degree* 5*.*

The absence of a morphism of degree 5 to $\mathbb{P}^1$ was already known at 21 of these levels [11, §7.2 and Table 2]; it is new at $74, 77, 79, 85, 88, 96, 98, 100, 103$. The finiteness of the points of degree 5 is new at all thirty levels.

**Theorem B.** *Exactly* 43 *strict intermediate curves $X_\Delta(N)$ have infinitely many points of degree* 5*. They are listed in Table 4, and each of them has a morphism of degree* 5 *to $\mathbb{P}^1$ over $\mathbb{Q}$.*

**Corollary E.** *The strict intermediate curves of minimum density degree* 5 *are exactly the four $\mathbb{Q}$-pentagonal curves of Orlić* [37, Theorems 1.4–1.5]*: $X_\Delta(31)$ with $\Delta = \langle -1, 2\rangle$ and $\Delta = \langle -1, 5\rangle$, $X_\Delta(44)$ with $\Delta = \langle -1, 5\rangle$, and $X_\Delta(125)$ with $\Delta = \langle -1, 4\rangle$.*

This parallels the case of $X_0(N)$, where $X_0(109)$ is the only curve of minimum density degree 5 [11].

1.3. **Methods.** We have tried to keep the new mathematical input to a minimum, so that the paper can be checked by a human reader. Wherever a published result or an inequality settles a case, we use it: Castelnuovo–Severi, the classifications of curves of small gonality [24, 25, 37], and the classifications of curves with infinitely many points of degree at most 4 [25, 6, 13, 11]. Everything else is done by three methods, each stated once:

I. *Explicit pencils*: an explicit divisor $D$ with $\ell(D) \geq 2$ gives $\mathrm{gon}_{\mathbb{Q}} \leq \deg D$ (§3.1).
II. *Finite-field exclusion*: at a prime of good reduction, a complete search over $\mathbb{F}_p$ shows that there is no function of degree at most $d$ (§3.2). We use the gonality algorithm of Derickx and Terao [15] and their package CurveArith. For the large curves $X_1(N)$ we keep their first phase (places and expansions of differentials) unchanged and replace their second phase by a search over a smaller family of divisors, run in compiled code (§3.3). For $X_1(48)$ we search instead on the quotient by a diamond involution (§5.1).
III. *Lifting arguments*: when a finite field allows more than $\mathbb{Q}$ does, a Mordell–Weil or local argument decides which classes over $\mathbb{F}_p$ come from $\mathbb{Q}$. This is needed for Theorems A and B and for one curve in Theorem C.

All computations were done in Magma V2.29-4 [5] and SageMath 10.7. The code, data and logs that verify all computations in this paper can be found on

`https://github.com/marin-varivoda/intermediate-modular-curves`;

§8 describes how to rerun them.

## 2. Preliminaries

2.1. **Curves and models.** The groups $\Delta$ and the curves $X_\Delta(N)$ are enumerated with Derickx's package MD Sage [9] (`intermediate_modular_groups`), as in [37, 13]. We identify each curve with its LMFDB label by checking that the corresponding open subgroup of $\mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$,

$$H_\Delta = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z}) : a \in \Delta \right\},$$

is conjugate to the group of the label. Rational Fourier expansions at the cusp $\infty$ describe the model of $X_\Delta(N)$ that parametrizes embeddings $\mu_N \hookrightarrow E$ up to $\Delta$; on this model the cusp $\infty$ is a rational point. This model is $\mathbb{Q}$-isomorphic to the one parametrizing points of order $N$ up to $\Delta$: quotienting by the marked subgroup and using the Weil pairing interchanges the two moduli problems over $\mathbb{Q}$ and carries the diamond operator $\langle a \rangle$ to $\langle a^{-1} \rangle$, which preserves the subgroup $\Delta$, so the isomorphism descends to $X_\Delta(N)$. On the second model the cusp $\infty$ corresponds to the cusp 0. We only use that $X_\Delta(N)$ has a rational cusp, which we call $\infty$.

For genus $g \geq 5$ we use the canonical model computed by MD Sage (`vanishing_quadratic_forms` with `minimize_strategy="LLL"`); all lower bounds over finite fields for Theorems C and D are computed on these models or on the models of $X_1(N)$ below (for Theorem A, on Magma's canonical models of $X_0(N)$). For $X_1(N)$ we use the plane models of Sutherland and van Hoeij [40]: each comes with explicit Tate normal form data $(E, P)$ over its function field with $P$ of order $N$, which gives the birational map to $X_1(N)$ (compare [41]).

2.2. **Good reduction.** All lower bounds over $\mathbb{Q}$ in this paper come from a prime of good reduction.

**Lemma 2.1** (Specialization)**.** *Let $C/\mathbb{Q}$ be a smooth projective geometrically integral curve, and let $\mathcal{C}$ be a smooth proper model of $C$ over $\mathbb{Z}_{(p)}$. Then $\mathrm{gon}_{\mathbb{Q}} C \geq \mathrm{gon}_{\mathbb{F}_p} \mathcal{C}_{\mathbb{F}_p}$. More generally, if $L$ is a line bundle on $C$ with $h^0(L) \geq 2$, then its extension to $\mathcal{C}$ has $h^0 \geq 2$ on $\mathcal{C}_{\mathbb{F}_p}$.*

*Proof.* The second statement is upper semicontinuity of $h^0$ [20, III.12.8], applied to the extension of $L$ (which exists because $\mathcal{C}$ is regular). If $f\colon C \to \mathbb{P}^1$ has degree $d$, take $L = f^*\mathcal{O}(1)$: its extension has degree $d$ and at least two sections on the special fibre, so $\mathrm{gon}_{\mathbb{F}_p} \mathcal{C}_{\mathbb{F}_p} \leq d$. This is standard; see for example [38] and [34]. □

We certify good reduction at $p$ in one of two ways.

(1) For a canonical model of genus $g$, we check that its reduction modulo $p$ is integral, has geometric genus $g$, and has a nonsingular $\mathbb{F}_p$-point. Good reduction then follows from [2, Lemma 2.3]. The script `good_reduction.m` verifies these conditions for every canonical model and prime used in this paper.
(2) For $X_1(N)$ we use the plane models $F(x, y) = 0$ of Sutherland [40], as Derickx and van Hoeij do [16]. Lemma 2.3 of [2] does not apply in this case, because these models are singular. Instead, the script `model_checks.m` checks that the reduction of $F$ modulo $p$ has the same degree in $y$ as $F$, that it is irreducible over $\mathbb{F}_p(x)$, and that its function field $K_p$ has exact constant field $\mathbb{F}_p$ and genus $g = g(X_1(N))$. Let $\mathcal{X}$ be the smooth proper model of $X_1(N)$ over $\mathbb{Z}_{(p)}$, which exists because $p \nmid N$ [8]. Because $\bar{F}$ is irreducible over $\mathbb{F}_p(x)$ of the same degree in $y$, the Gauss valuation of $\mathbb{Q}(x)$ extending the $p$-adic valuation has a unique extension to $\mathbb{Q}(x, y)$, with residue field $K_p$, so it is the function field of an irreducible component of the special fibre of a normal proper model $\mathcal{Y}$ of $X_1(N)$ over $\mathbb{Z}_{(p)}$ (the normalization of the closure of the plane curve). Let $\mathcal{Y}' \to \mathcal{Y}$ be a desingularization [32, §8.3.4]; every component of $\mathcal{Y}_{\mathbb{F}_p}$ is birational to a component of $\mathcal{Y}'_{\mathbb{F}_p}$. The smooth model $\mathcal{X}$ is relatively minimal (its special fibre is irreducible of genus $g$, so contains no exceptional curve), hence, as $g \geq 1$, it is the minimal regular model, and every regular model dominates it [32, 9.3.21 and §9.3]. So the birational map $\mathcal{Y}' \dashrightarrow \mathcal{X}$ is a morphism, and by the factorization theorem [32, §9.2.1] it is a composition of blow-ups of closed points. Hence the components of $\mathcal{Y}'_{\mathbb{F}_p}$ are the strict transform of $X_1(N)_{\mathbb{F}_p}$ and exceptional components, of geometric genus 0. As $K_p$ has genus $g \geq 2$, it is the function field of $X_1(N)_{\mathbb{F}_p}$.

### 2.3. **Tools from the geometry of curves.**

**Lemma 2.2** (Castelnuovo–Severi [29, Theorem 14])**.** *Let $C$ be a curve of genus $g$ over a perfect field $k$, with morphisms $C \to C_1$ and $C \to C_2$ over $k$ of degrees $d_1, d_2$ to curves of genera $g_1, g_2$. If $g > d_1 g_1 + d_2 g_2 + (d_1 - 1)(d_2 - 1)$, then both morphisms factor through a common morphism $C \to C'$ of degree $> 1$ defined over $k$.*

We use it mainly with $C_2 = \mathbb{P}^1$: if $C$ has a morphism of degree $m$ to a curve of genus $h$ and a morphism to $\mathbb{P}^1$ of degree $d$ with $\gcd(d, m) = 1$, then a common factor of degree $> 1$ is impossible, so $g \leq mh + (m - 1)(d - 1)$.

**Lemma 2.3** (Pullback)**.** *If $\pi\colon C \to D$ has degree $m$ over $\mathbb{Q}$, then $\operatorname{gon}_{\mathbb{Q}} C \leq m \cdot \operatorname{gon}_{\mathbb{Q}} D$.*

**Lemma 2.4** (Hilbert irreducibility)**.** *If a curve $C/\mathbb{Q}$ has a morphism $C \to \mathbb{P}^1$ of degree $d$ over $\mathbb{Q}$, then it has infinitely many points of degree $d$.*

*Proof.* By Hilbert's irreducibility theorem, the set of $t \in \mathbb{P}^1(\mathbb{Q})$ whose fibre is a single closed point of degree $d$ is the complement of a thin set, and in particular infinite. □

**Lemma 2.5** (Frey [19])**.** *If $C/\mathbb{Q}$ has infinitely many points of degree $d$, then $\operatorname{gon}_{\mathbb{Q}} C \leq 2d$.*

**Lemma 2.6** (Abramovich [1])**.** *A modular curve of index $\mu$ in $\mathrm{PSL}_2(\mathbb{Z})$ has $\operatorname{gon}_{\mathbb{C}} \geq \frac{\lambda_1}{24}\mu$, where $\lambda_1$ is a lower bound for the smallest positive eigenvalue of the Laplacian on congruence subgroups. By Kim and Sarnak* [30, Appendix 2] *one may take $\lambda_1 = \frac{975}{4096}$.*

Since $\frac{119}{12000} < \frac{975}{24 \cdot 4096}$, a modular curve with $119\mu > 120000$ has $\operatorname{gon}_{\mathbb{C}} \geq \frac{975}{24 \cdot 4096}\mu > \frac{119\mu}{12000} > 10$, so by Lemma 2.5 it has only finitely many points of degree 5; this is the form used in [37, Theorem 3.16] and below.

## 3. Three methods

Throughout, $C$ is a smooth projective geometrically integral curve of genus $g$ over a field $k$, and for a divisor $D$ we write $L(D) = \{f \in k(C)^\times : \operatorname{div} f + D \geq 0\} \cup \{0\}$ and $\ell(D) = \dim L(D)$. A nonconstant $f \in k(C)$ has degree $\deg f = \deg(f)_\infty$, the degree of its pole divisor, and $\operatorname{gon}_k C \leq d$ if and only if there is an effective divisor $D$ of degree at most $d$ with $\ell(D) \geq 2$.

3.1. **Method I: explicit pencils.** To prove $\operatorname{gon}_{\mathbb{Q}} C \leq d$ we exhibit a rational divisor $D$ of degree $d$ and compute $\ell(D) \geq 2$ with Magma's Riemann–Roch routine. The divisor need not be effective: for $0 \neq f \in L(D)$ the divisor $D + \operatorname{div} f$ is effective, rational and linearly equivalent to $D$. The divisors used are of three kinds:

(1) divisors supported on rational cusps, or on cusps and other low-degree points, whose Riemann–Roch spaces we compute from $q$-expansions at all cusps, using Zywina's package [46];
(2) pullbacks along a forgetful map $X_\Delta(N) \to X_{\Delta'}(M)$ of a pencil of minimal degree on the target, whose gonality is known (Lemma 2.3);
(3) low genus: if $g \geq 2$ and $P \in C(\mathbb{Q})$, then $\ell(K - (g-2)P) = \ell((g-2)P) + 1 \geq 2$ by Riemann–Roch, so $\operatorname{gon}_{\mathbb{Q}} C \leq g$; if $g \leq 1$ and $P \in C(\mathbb{Q})$, then $\ell(2P) \geq 2$ (and $\ell(P) = 2$ if $g = 0$).

For $X_1(N)$ the upper bounds in Theorem D are the degrees of the modular units of Derickx and van Hoeij [16, Table 1].

3.2. **Method II: finite-field exclusion.** To prove $\operatorname{gon}_{\mathbb{Q}} C > d$, we choose a prime $p$ of good reduction (Lemma 2.1) and show that $C_{\mathbb{F}_p}$ has no nonconstant function of degree at most $d$. For this we use the algorithm of Derickx and Terao [15], implemented in their Magma package CurveArith; it works over any finite field $\mathbb{F}_q$ (here $q = p$) and has two phases.

*Phase 1 (places and expansions).* Let $N_1 = \#C(\mathbb{F}_q)$ and $n = \lceil N_1/(q+1) \rceil$. CurveArith enumerates the places of $C_{\mathbb{F}_q}$ of degree at most $d - \min(n, d-1)$, fixes a basis $\omega_1, \dots, \omega_g$ of $H^0(C, \Omega^1)$, and at each place $P$ computes the first $m_P$ coefficients of the expansions of the $\omega_i$ in a uniformizer at $P$, as vectors over $\mathbb{F}_q$ ($\deg P$ coordinates per coefficient). The precision is $m_P = d$ at rational places and $m_P = \lfloor (d - \min(n, d-1))/e \rfloor$ at places of degree $e \geq 2$; this suffices for all divisors of degree $d$ containing at least $\min(n, d-1)$ rational places. For an effective divisor $D = \sum a_P P$ with $a_P \leq m_P$ this gives a $g \times \deg D$ matrix $M_D$, whose columns are the coefficients of index $< a_P$ at each $P$ in its support.

**Lemma 3.1** ([15])**.** *$\ell(D) = \deg D + 1 - \operatorname{rank} M_D$.*

*Proof.* By Riemann–Roch, $\ell(D) = \deg D + 1 - g + \ell(K - D)$, and $\ell(K - D) = g - \operatorname{rank} M_D$ is the dimension of the space of holomorphic differentials vanishing on $D$. □

**Lemma 3.2** ([15, 16], [34, Lemma 3.1])**.** *If $C_{\mathbb{F}_q}$ has a nonconstant function $f$ of degree at most $d$, it has one whose pole divisor contains at least $n$ distinct rational places.*

*Proof.* The $N_1$ rational points are distributed among the $q+1$ fibres of $f$ over $\mathbb{P}^1(\mathbb{F}_q)$, so one fibre contains at least $n$ of them. Compose $f$ with an automorphism of $\mathbb{P}^1$ over $\mathbb{F}_q$ moving that fibre to $\infty$. □

*Phase 2 (search).* CurveArith tests all effective divisors $D$ of degree $d$ containing at least $n$ rational places for $\ell(D) \geq 2$ (Lemma 3.1). This is enough for every strict intermediate curve in this paper. For the curves $X_1(N)$ with $41 \leq N \leq 50$ there are far too many such divisors, and we use instead a smaller family of larger divisors, as follows.

3.3. **A smaller search family.** Let $S = \{P_1, \dots, P_{N_1}\}$ be the rational places, and also write $S = \sum P_i$. Fix an integer $b \geq 1$. For an effective divisor $Q$ with no rational place in its support and an effective divisor $E$ supported on $S$, put $s = \#\operatorname{supp} E$ and

$$c(E, Q) = b\, s + \max(n - s, 0) + \deg E + \deg Q. \tag{1}$$

Let $\mathcal{F}_b(d)$ be the set of divisors $H = bS + E + Q$ with $\deg Q \leq d - n$, $c(E, Q) \leq d$, and $E$ maximal with this property for its $Q$ (for the order $E \leq E'$).

**Proposition 3.3.** *Suppose $n \geq 1$. If $C_{\mathbb{F}_q}$ has a nonconstant function of degree at most $d$, then there is $H \in \mathcal{F}_b(d)$ such that $L(H)$ contains a nonconstant function of degree at most $d$.*

*Proof.* By Lemma 3.2 there is such an $f$ whose pole divisor $D = A + Q$ has at least $n$ distinct rational places in its support, where $A$ is supported on $S$ and $Q$ on the other places. Put $E = \sum_{P \in S} \max(v_P(A) - b, 0)P$ and $s = \#\operatorname{supp} E$. A place in $\operatorname{supp} E$ contributes $b + v_P(E)$ to $\deg A$, and $\operatorname{supp} A$ has at least $\max(n - s, 0)$ further places, each contributing at least 1. Hence $c(E, Q) \leq \deg A + \deg Q = \deg f \leq d$, and in particular $\deg Q \leq d - n$. Since $v_P(A) \leq b + v_P(E)$ for every $P \in S$, we have $D \leq bS + E + Q$. Enlarging $E$ to a maximal $E'$ with $c(E', Q) \leq d$ keeps $D \leq bS + E' + Q$, so $f \in L(bS + E' + Q)$ with $bS + E' + Q \in \mathcal{F}_b(d)$. □

The divisors in $\mathcal{F}_b(d)$ have degree larger than $d$, so $\ell(H) \geq 2$ is not enough: we need the least degree of a nonconstant function in $L(H)$. It can be read off from the same matrix $M_H$.

**Lemma 3.4.** *Let $H$ be effective. The space $L(H)/k$ is isomorphic to the space of vectors $x$ with $M_H x = 0$. Under this isomorphism, the order of the pole of $f$ at $P \in \operatorname{supp} H$ is the largest $j \leq v_P(H)$ such that the coordinates of $x$ belonging to the coefficient of index $j - 1$ at $P$ are not all zero.*

*Proof.* For each $P \in \operatorname{supp} H$ let $V_P = \mathcal{O}_P/\mathfrak{m}_P^{v_P(H)}$ and $V = \bigoplus V_P$. At each $P$, CurveArith writes each $\omega_i$ as $g_i\, \eta_P$, where $\eta_P$ is one of the basis differentials that does not vanish at $P$, and expands $g_i$ in a uniformizer $u_P$; the $i$th row of $M_H$ is the coordinate vector (over $k$, using a $k$-basis of the residue field at $P$) of the jet of $g_i$ in $V$, in which, for every $j$, the subspace $\mathfrak{m}_P^j/\mathfrak{m}_P^{v_P(H)}$ is cut out by the vanishing of the coordinates of index $< j$; indeed, CurveArith computes the coefficients one at a time, subtracting a lift of each coefficient and dividing by the uniformizer, and this is $k$-linear. The pairing $(\phi, h) \mapsto \sum_P \operatorname{Tr}_{k(P)/k} \operatorname{Res}_P(\phi\, h\, \eta_P)$, which is perfect because $\eta_P$ generates the differentials at $P$, identifies the principal parts $\bigoplus_P \mathfrak{m}_P^{-v_P(H)}/\mathcal{O}_P$ with the dual of $V$, a principal part having pole order $\leq j$ at $P$ exactly when it annihilates $\mathfrak{m}_P^j/\mathfrak{m}_P^{v_P(H)}$. By the residue theorem and Serre duality, a principal part is the principal part of a function in $L(H)$ (unique up to constants) if and only if it annihilates the image of $H^0(C, \Omega^1)$ in $V$, that is, if and only if its coordinate vector $x$ satisfies $M_H x = 0$. □

*Precision.* The expansions computed in Phase 1 suffice for every $H \in \mathcal{F}_b(d)$: at a rational place $P \in \operatorname{supp} E$ we have $v_P(H) = b + v_P(E) \leq bs + \deg E \leq d$, which is CurveArith's precision there, and at a place $P$ of degree $e \geq 2$ we have $e\, v_P(H) \leq \deg Q \leq d - n$, so $v_P(H) \leq \lfloor (d - \min(n, d - 1))/e \rfloor$, which is CurveArith's precision there. (The program checks this before searching.)

So for $H \in \mathcal{F}_b(d)$ we compute the kernel of $M_H$ and check whether some nonzero kernel vector, up to scalars, gives a pole divisor of degree at most $d$. By Proposition 3.3, if this fails for every $H \in \mathcal{F}_b(d)$, then $\operatorname{gon}_{\mathbb{F}_q} C > d$. The same argument with $b = 0$ and all effective divisors of degree at most $d$ handles curves without rational points; we did not need this case.

*Remark* 3.5. Proposition 3.3 and Lemma 3.4 are an elementary variation on the second phase of [15]; the first phase is used unchanged. The point is that $\mathcal{F}_b(d)$ is much smaller than the set of

divisors of degree $d$ with $n$ rational places: the rational part is fixed to $bS$ up to the small excess $E$, and the kernels of $M_H$, of dimension $\ell(H)-1$, are small, because the divisors $H$ have degree only slightly above $d$. For a fixed $Q$, the matrices $M_{bS+E+Q}$ share the part belonging to $bS+Q$, so we eliminate it once and add the part belonging to $E$ incrementally. This variation was first implemented as an option `Al:="Cover"` of CurveArith's `CAHasFunctionOfDegreeAtMost`, as a patch to CurveArith (`certificates/D_x1_gonality/curvearith_cover_option.patch`, which applies to CurveArith commit `0263b34` and adds its own tests; the certificates use the unmodified CurveArith); for the curves $X_1(N)$ we run the same search in a short C++ program (§8).

3.4. **Method III: lifting.** When $C_{\mathbb{F}_p}$ has more functions or divisor classes of a given kind than can come from $\mathbb{Q}$, we use the structure of $J(\mathbb{Q})$, where $J=\operatorname{Jac} C$, to decide which of them lift. The arguments are of three types, each explained where it is used:

(1) *rank zero*: if $J(\mathbb{Q})$ is finite, reduction at an odd good prime is injective on $J(\mathbb{Q})$, and a finite list of classes over $\mathbb{F}_p$ can be compared with explicit rational classes (§6, §7);
(2) *a known positive-rank part*: if $J(\mathbb{Q})=A(\mathbb{Q})+T$ for an explicit abelian subvariety $A$ and an explicit finite group $T$, the reduction of $J(\mathbb{Q})$ is contained in an explicit subgroup of $J(\mathbb{F}_p)$ (the curve 43.132.9.$a$.2 in §4, and Route Q in §6, Table 9);
(3) *local rigidity*: if over $\mathbb{F}_p$ the relevant locus agrees, to first order, with a known family defined over $\mathbb{Q}$, a formal implicit function argument shows that it agrees over $\mathbb{Z}_p$ (Lemma 6.2).

## 4. Gonality of intermediate curves: proof of Theorem C

There are 76 strict intermediate curves with $N\le 50$, of which 52 have $N\le 40$; for the 39 of these of genus $\ge 3$ both bounds are in [37, Theorem 1.2, Table 1], the other 13 have genus $\le 1$, and we only recall them in Table 3. The remaining 24 curves, with $41\le N\le 50$, are treated as follows.

4.1. **Upper bounds.** Each upper bound is Method I.

- *Pullbacks.* Fourteen curves map, by a forgetful map of degree $m$, to a curve whose gonality $\gamma$ is known, with $m\gamma$ equal to the lower bound. The targets have genus at most 3, or are hyperelliptic, or are curves $X_1(M)$ with $M\le 40$ [16]; the degree and target of each map are listed in the data file of the accompanying repository.
- *Cuspidal divisors.* For five curves (41.168.11.$b$.1, 41.210.11.$b$.1, 41.420.21.$d$.1, 45.432.21.$g$.2, 49.392.19.$i$.2), a rational divisor $D$ supported on the cusps has degree equal to the lower bound and $\ell(D)\ge 2$. We compute $\ell(D)=\deg D+1-g+\ell(K-D)$, where $H^0(K-D)$ is the space of weight-2 cusp forms vanishing to the prescribed orders at the cusps, from $q$-expansions at all cusps computed with Zywina's package [46]. Conditions at one cusp of each Galois orbit suffice, since the basis is rational.
- *Published.* 44.144.8.$c$.4 is in Orlić's list of $\mathbb{Q}$-pentagonal curves [37, Theorems 1.4–1.5], and 45.144.5.$a$.1 in the list of $\mathbb{Q}$-tetragonal curves [37, Theorem 1.3]; 49.168.3.$b$.1 is a nonhyperelliptic genus-3 curve with a rational point.
- *Two special divisors at level* 43. For 43.308.15.$b$.1 (genus 15) we take $D=3C_3+C_4+P_5$, where $C_3,C_4$ are rational cusps and $P_5$ is a point of degree 5, the image of a point of degree 15 on $X_1(43)$ found by van Hoeij [43]; then $\deg D=9$ and $\ell(D)=2$. The three points are given by explicit coordinates on the canonical model, and $\ell(D)=2$ is checked there: the conditions for a canonical section to vanish on $D$ have rank 8, so $\ell(K-D)=7$ and $\ell(D)=2$ by Riemann–Roch. For 43.132.9.$a$.2 (genus 9), let $\pi\colon X\to X_0^+(43)=E$ be the map of degree 6 to the elliptic curve 43$a$1; with rational cusps $C_1,C_2,C_3$, the

divisor $D = 3C_3 + 7C_2 - 3C_1 + \pi^*(G) - \pi^*(O)$ for a point $G \in E(\mathbb{Q})$ of infinite order has degree 7 and $\ell(D) = 2$.

4.2. **Lower bounds.** Twenty-one of the 24 lower bounds are Method II, using CurveArith's `CAGonality` unchanged at a prime of good reduction (Table 3). Six of these curves (41.84.5.$a$.1, 44.144.8.$c$.4, 45.144.5.$a$.1 and the three curves of genus 5 and 7 at level 48) are also covered by Orlić's classification [37]; we recompute them as a check. The longest computation, for 48.384.19.$bc$.1 at $p = 7$, takes about an hour; all others take at most eleven minutes. The curves 49.168.3.$b$.1 and 50.180.4.$a$.2 have genus 3 and 4 and are not hyperelliptic [25, Theorem 3.6], hence have gonality at least 3.

The remaining curve, 43.132.9.$a$.2, needs Method III.

As a consistency check, all 24 values lie within the bounds for the $\mathbb{Q}$-gonality recorded in the LMFDB [42], which gives the exact value for six of them, in agreement with Table 3 (`lmfdb_crosscheck.py` in the repository).

4.3. **The curve** 43.132.9.$a$.2**.** Let $X = X_\Delta(43)$ with $\Delta = \langle -1, 2\rangle = \{\pm1, \pm2, \pm4, \pm8, \pm11, \pm16, \pm21\}$. It has genus 9, three rational cusps $C_1, C_2, C_3$ and one cusp orbit $C_{(3)}$ of degree 3. The forgetful map $X \to X_0(43)$ has degree 3, and composing with the quotient by the Atkin–Lehner involution gives $\pi\colon X \to X_0^+(43) = E$, of degree 6, where $E$ is the elliptic curve 43$a$1 of rank 1.

**Proposition 4.1.** $\operatorname{gon}_{\mathbb{Q}} X = 7$.

The Castelnuovo–Severi inequality does not help here: the maps to $X_0(43)$ (genus 3, degree 3) and to $E$ (degree 6) exclude nothing in degrees 2 to 6. Over $\mathbb{F}_2$ the gonality is 5 and over $\mathbb{F}_5$ it is 4; we use $p = 3$, where $X$ has good reduction.

*Proof.* The upper bound is the divisor above. For the lower bound, let $J = \operatorname{Jac} X$, and let $c = C_3$ be the base point of the Abel–Jacobi map.

*Step 1: over* $\mathbb{F}_3$. `CAGonality` gives $\operatorname{gon}_{\mathbb{F}_3} X_{\mathbb{F}_3} = 6$, so by Lemma 2.1 there is no function of degree at most 5 over $\mathbb{Q}$. Here $n = \lceil 4/4\rceil = 1$, and testing the 845 effective divisors of degree 6 over $\mathbb{F}_3$ containing a rational place shows that exactly four have $\ell \geq 2$, and that they are linearly equivalent. These four divisors exhaust $W_6^1(X_{\mathbb{F}_3})(\mathbb{F}_3)$: a class in it has a pencil without base points (a base point would give a function of degree at most 5), hence a morphism to $\mathbb{P}^1$ of degree 6 over $\mathbb{F}_3$, and by Lemma 3.2 with $n = 1$ one of its fibres over $\mathbb{P}^1(\mathbb{F}_3)$ is an effective divisor in the class that contains a rational place. So $W_6^1(X_{\mathbb{F}_3})(\mathbb{F}_3)$ is a single class $T$, and $T - 6c$ has order 1596 in $J(\mathbb{F}_3)$, a group of order 19152.

*Step 2: the rational points of* $J$. Let $A = \pi^* E \subseteq J$ and $B = J/A$. The pullback $\pi^*$ is injective: its kernel is a rational finite subgroup of $E$ of order dividing $\deg \pi = 6$, hence cyclic since 6 is squarefree, and a nontrivial one would give a rational cyclic isogeny of 43$a$1, which has none. So $A(\mathbb{Q}) \cong E(\mathbb{Q}) \cong \mathbb{Z}$. The abelian variety $B$ is isogenous to a product of modular abelian varieties of dimensions 2, 2 and 4 whose $L$-values at 1 are nonzero (exact modular-symbol computations of $L$-ratios 2/7, 3/16, 1/361); by Kato [27, Corollary 14.3], $B(\mathbb{Q})$ is finite. Its order divides $\gcd_p \#B(\mathbb{F}_p)$ over odd primes $p \neq 43$, since reduction is injective on rational torsion at such primes [28, Appendix]; the gcd over $p = 3, 5, 23$ is already 532. (Point counts are invariant under isogeny, so we may compute them on the modular abelian variety isogenous to $B$.) On the other hand, the subgroup $\mathcal{C} \subseteq J(\mathbb{Q})$ generated by $[C_2 - C_1]$, $[C_3 - C_1]$ and $[C_{(3)} - 3C_3]$ is finite (its generators are torsion, as checked below; this is also an instance of the theorem of Manin and Drinfeld [33, 17]), of order at least 532 (its reduction modulo 3 has order 532), and it meets $A(\mathbb{Q})$ trivially, as $A(\mathbb{Q})$ is torsion-free. So $\mathcal{C}$ maps isomorphically onto $B(\mathbb{Q})$, and

$$J(\mathbb{Q}) = A(\mathbb{Q}) + \mathcal{C}.$$

*Step 3: conclusion.* Let $H_3 \subseteq J(\mathbb{F}_3)$ be the subgroup generated by $\pi^*E(\mathbb{F}_3)$ and the reduction of $\mathcal{C}$. It has order 3192, and $T - 6c \notin H_3$. If $f\colon X \to \mathbb{P}^1$ had degree 6 over $\mathbb{Q}$, the class of $L = f^*\mathcal{O}(1)$ would satisfy $[L - 6c] \in J(\mathbb{Q})$, so its reduction would lie in $H_3$; but by Lemma 2.1 the reduction of $L$ is a degree-6 class with two sections, that is $T$. This contradiction shows $\operatorname{gon}_{\mathbb{Q}} X \geq 7$. □

*Remark* 4.2. The saturation in Step 2 cannot be avoided: $J(\mathbb{F}_3)$ has cyclic 3-part, and the 3-component of $T - 6c$ lies in $\pi^*E(\mathbb{F}_3)$, so $T - 6c \notin H_3$ only because of its 2-, 7- and 19-parts. In particular the coarser statement "$532 \cdot J(\mathbb{Q}) \subseteq A(\mathbb{Q})$" would not suffice.

The certificate for Proposition 4.1 runs in about 12 seconds; the computations of $L$-ratios and of $\#B(\mathbb{F}_p)$ take a few minutes. The scripts work with the LMFDB model of $X$, its map to $X_0(43)$ and its cusps; the script `identify_model.m` shows that this model is isomorphic over $\mathbb{Q}$ to the MD Sage model, by an explicit linear change of coordinates (found by matching points over $\mathbb{F}_{101}$ and $\mathbb{F}_{103}$, and checked exactly). The three classes generating $\mathcal{C}$ are certified to be torsion directly over $\mathbb{Q}$ (`torsion_classes.m`): with $D_1 = C_2 - C_1$, $D_2 = C_3 - C_1$ and $E = C_{(3)} - (C_1 + C_2 + C_3)$, the divisors $38D_1$, $38D_2$ and $7E$ are principal, and $C_{(3)} - 3C_3 = E + D_1 - 2D_2$. So the proof does not depend on knowing that these points are cusps.

## 5. Gonality of $X_1(N)$: proof of Theorem D

*Proof of Theorem D. Upper bounds.* Derickx and van Hoeij [16, Table 1] give, for each $N$, a modular unit on $X_1(N)$ over $\mathbb{Q}$ whose degree is the value in Theorem D.

*Lower bounds.* Let $u$ be the value in Theorem D and $d = u - 1$. For $N = 48$ see §5.1. For the other $N$ we apply Method II to $X_1(N)$ at the prime $p$ of Table 1, which does not divide $N$; §2 explains why the reduction of the plane model of [40] computes $X_1(N)_{\mathbb{F}_p}$, and we check its genus. Phase 1 is CurveArith's: the places of degree at most $R = d - \min(n, d-1)$ and the expansions of a basis of holomorphic differentials at them, with CurveArith's precisions. Phase 2 is the search of §3.3 over the family $\mathcal{F}_b(d)$. In every case it finds no function of degree at most $d$. By Proposition 3.3 and Lemma 3.4, $\operatorname{gon}_{\mathbb{F}_p} X_1(N)_{\mathbb{F}_p} > d$, and by Lemma 2.1, $\operatorname{gon}_{\mathbb{Q}} X_1(N) \geq u$. □

Table 1. Theorem D: Method II for $X_1(N)$. $n = \lceil N_1/(p+1) \rceil$; places up to degree $R = d - \min(n, d-1)$; $b$ the baseline; $\#\mathcal{F}_b(d)$ the number of divisors searched. Time: Phase 1 (its longest process, on a shared machine) + Phase 2.

| $N$ | $g$ | $p$ | $d$ | $N_1$ | $n$ | $R$ | places | $b$ | $\#\mathcal{F}_b(d)$ | time |
|---|---|---|---|---|---|---|---|---|---|---|
| 41 | 51 | 2 | 21 | 20 | 7 | 14 | 2669 | 2 | 2643150 | 171 min + 14 s |
| 42 | 25 | 5 | 11 | 12 | 2 | 9 | 277397 | 1 | 3860784 | 648 min + 153 s |
| 43 | 57 | 2 | 23 | 21 | 7 | 16 | 8515 | 2 | 17156720 | 367 min + 50 s |
| 44 | 36 | 3 | 14 | 15 | 4 | 10 | 9518 | 1 | 3133892 | 34 min + 7 s |
| 45 | 41 | 2 | 17 | 12 | 4 | 13 | 1608 | 2 | 265608 | 86 min + 1 s |
| 46 | 45 | 3 | 18 | 22 | 6 | 12 | 69995 | 1 | 374417725 | 884 min + 950 s |
| 47 | 70 | 2 | 28 | 23 | 8 | 20 | 110354 | 2 | 651609964 | 1813 min + 3565 s |
| 49 | 69 | 2 | 20 | 21 | 7 | 13 | 1497 | 2 | 2167431 | 207 min + 13 s |
| 50 | 48 | 3 | 14 | 20 | 5 | 9 | 3369 | 2 | 41835 | 161 min + 1 s |

The second phase also counts the divisors it tests and compares the total with an independent count of $\#\mathcal{F}_b(d)$ computed from the numbers of places of each degree; they agree in every case. As a check of the whole pipeline, the same programs reproduce $\operatorname{gon}_{\mathbb{F}_2} X_1(29) = 11$ (degree 10 is excluded after testing 8568 divisors, and a function of degree 11 is found). For

$X_1(47)$, an earlier run with a different Phase 1 driver (a different distribution of the places among the processes; in both runs the expansions are computed by CurveArith) and an earlier build of the Phase 2 program tested exactly the same number 651,609,964 of divisors and also found no function of degree at most 28. It ran on the same machine and with the same version of Magma, so it checks our drivers rather than the platform; its record is in `certificates/D_x1_gonality/runs/x1_47_independent`.

5.1. **The level 48: a double cover.** For $N = 48$ Method II over $\mathbb{F}_5$ or $\mathbb{F}_7$ would need all places of degree up to 12, which is out of reach. We use instead the quotient of $X = X_1(48)$ by the diamond operator $\langle 23\rangle$, an involution; the quotient is $Y = X_{\{\pm1,\pm23\}}(48)$, of genus 13, and $X$ has genus 37. Everything is computed on $Y$.

Let $k$ be a finite field of odd characteristic, and $\pi\colon X \to Y$ a morphism of degree 2 of smooth curves over $k$, the quotient by an involution $\sigma$. For a function $a$ on $Y$ write $\pi^*a = a\circ\pi$; these are exactly the $\sigma$-invariant functions on $X$. Fix a function $w$ on $X$ with $\sigma w = -w$; then $w^2 = \pi^*\delta$ for a function $\delta$ on $Y$. Write $\operatorname{div}_Y \delta = B - 2H$ with $B$ the sum of the branch points, so that $c := \deg H = \frac{1}{2}\deg B = g_X - 2g_Y + 1$. (For a hyperelliptic curve $w^2 = \delta(x)$ over $Y = \mathbb{P}^1$, all of this is classical: every function is $a(x) + b(x)w$.)

**Lemma 5.1.** (1) *Every function $f$ on $X$ is uniquely $f = \pi^*a + \pi^*b\cdot w$ with $a, b$ functions on $Y$, and $f$ comes from $Y$ if and only if $b = 0$.*

(2) *Put $t = 2a$ and $n = a^2 - b^2\delta$. Then $f + \sigma f = \pi^*t$, $f\cdot\sigma f = \pi^*n$, and for every place $P$ of $Y$ the total pole order of $f$ above $P$ is $\deg P\cdot\max(0, -v_P(t), -v_P(n))$. In particular $\deg_X f \geq \deg_Y n$ and $\deg_X f \geq \deg_Y t$.*

(3) *If $\deg_X f = d$ and $b \neq 0$, there are an effective divisor $E$ on $Y$ of degree $d - c$ and $u \neq 0$, $h$ in $L_Y(E + H)$ with $f = (\pi^*h + w)/\pi^*u$.*

*Proof.* (1) Take $\pi^*a = (f + \sigma f)/2$ and $\pi^*b = (f - \sigma f)/(2w)$, both $\sigma$-invariant. (2) The first two identities are direct. Measure the orders of the two values of $f$ above $P$ in units of $v_P$, say $\alpha, \beta$ (if a single ramified point $Q$ lies above $P$, then $\alpha = \beta = v_Q(f)/2$; if a single unramified point $Q$ of degree $2\deg P$ lies above $P$, then $\alpha = \beta = v_Q(f)$). The pole order above $P$ is $\deg P(\max(0,-\alpha) + \max(0,-\beta))$, while $v_P(n) = \alpha + \beta$ and $v_P(t) \geq \min(\alpha,\beta)$ with equality if $\alpha \neq \beta$. If $\alpha, \beta \geq 0$ both sides vanish; if $\alpha < 0 \leq \beta$ then $v_P(t) = \alpha \leq v_P(n)$ and both sides are $-\alpha$; if $\alpha, \beta < 0$ then $v_P(n) < v_P(t)$ and both sides are $-\alpha - \beta$. (3) Since $\operatorname{div}_X w = R - \pi^*H$ with $R$ the ramification divisor, for every divisor $A$ on $Y$ the poles of $f$ are bounded by $\pi^*A$ if and only if $a \in L_Y(A)$ and $b \in L_Y(A - H)$. Take $A = \pi_*$ of the pole divisor of $f$, of degree $d$, and $E = \operatorname{div} b + A - H \geq 0$, $u = 1/b$, $h = a/b$. □

**Proposition 5.2.** *$X$ has no nonconstant function of degree at most $m$ if $2\operatorname{gon}_k Y > m$ and, for every effective divisor $E$ on $Y$ of degree $m - c$, every $u \neq 0$ and every $h$ in $L_Y(E + H)$, the function $(\pi^*h + w)/\pi^*u$ has degree $> m$.*

*Proof.* Functions with $b = 0$ are $\pi^*a$, of degree $2\deg_Y a$. For the others use Lemma 5.1(3) and enlarge $E$ by $m - d$ points, which only enlarges $L_Y(E + H)$. □

For $X_1(48)$ we take $k = \mathbb{F}_7$ and $m = 15$, so $c = 12$ and $E$ has degree 3. The diamond operator $\langle 23\rangle$ is defined over $\mathbb{Q}$ and extends to the smooth model of $X_1(48)$ over $\mathbb{Z}_{(7)}$; since 2 is invertible, the quotient commutes with base change, so $Y_{\mathbb{F}_7} = X_1(48)_{\mathbb{F}_7}/\langle 23\rangle$ is the reduction of $Y$. We use MD Sage's canonical model of $Y$ (coordinates $y_0, \ldots, y_{12}$, a basis of $S_2(\Gamma_Y)$ with integral $q$-expansions, $y_0 = q + O(q^2)$), checked to have good reduction at 7 as in §2. Let $\psi \in S_2(\Gamma_1(48))$ have integral $q$-expansion, nonzero modulo 7, with $\langle 23\rangle\psi = -\psi$, and put $w = \psi/y_0$, a function on $X_1(48)$ with $\sigma w = -w$. Then $\delta = w^2 = \psi^2/y_0^2$ is a ratio of two forms of weight 8 on $Y$, $\delta = \operatorname{Nm}(y)/\operatorname{Dn}(y)$, from an identity $\psi^2\operatorname{Dn}(y) = y_0^2\operatorname{Nm}(y)$ with integral coefficients checked to

the Sturm bound. The reduction $\mathrm{Dn}(\bar{y})$ is not identically zero on $Y_{\mathbb{F}_7}$ (the certificate divides by it), so $\bar{\delta} = \mathrm{Nm}(\bar{y})/\mathrm{Dn}(\bar{y})$ is a well-defined function on $Y_{\mathbb{F}_7}$, and $\psi^2 \equiv y_0^2\bar{\delta}$. Reducing modulo 7, $\bar{w}$ is a function on $X_1(48)_{\mathbb{F}_7}$ with $\sigma\bar{w} = -\bar{w}$ and $\bar{w}^2 = \bar{\delta}$, so the function field of $X_1(48)_{\mathbb{F}_7}$ is $\mathbb{F}_7(Y)(\sqrt{\bar{\delta}})$; as a check, Magma computes its genus as 37. `CAGonality` gives $\mathrm{gon}_{\mathbb{F}_7} Y = 8$, so $2\,\mathrm{gon}\, Y = 16 > 15$. There are 1328 effective divisors $E$ of degree 3 on $Y_{\mathbb{F}_7}$; each $L_Y(E+H)$ has dimension 3, giving 57 choices of $u$ up to scalars and 49 of $h$ modulo $u$, and 3,709,104 functions in all. The bound $\deg_X f \geq \deg_Y n$ of Lemma 5.1(2) excludes all but 18 of them, and for those the exact degree, computed on $Y$, is at least 16. Hence $\mathrm{gon}_{\mathbb{F}_7} X_1(48) > 15$, and $\mathrm{gon}_{\mathbb{Q}} X_1(48) = 16$ by Lemma 2.1. The divisors $E$ are split into parts by a key that depends only on $E$ (not on the order in which Magma lists places), so the parts cover every $E$ exactly once; the longest part took 12.5 hours. The degree formula of Lemma 5.1(2) was checked against Magma's degree computation on $X$ itself for a sample of the candidates, and the whole procedure was tested on $X_1(24)$ and $X_1(26)$, where it reproduces the known gonalities.

## 6. Points of degree 5 on $X_0(N)$: proof of Theorem A

Let $C = X_0(N)$ for one of the thirty levels of Theorem A, of genus $g$, and $J = \mathrm{Jac}\, C$. The rational cusp $\infty$ identifies $\mathrm{Pic}^5(C)$ with $J$. For $r \geq 0$ let $W_5^r \subseteq \mathrm{Pic}^5(C)$ be the locus of classes with $h^0 \geq r+1$. Table 2 lists, for every level, the argument for each of the two assertions of Theorem A.

### 6.1. **The framework.**

**Lemma 6.1.** *If $C$ has no function of degree 5 over $\mathbb{Q}$, then the points of degree 5 inject into $W_5^0(\mathbb{Q})$, and a point of degree 5 is the unique effective divisor in its class.*

*Proof.* If $D$ is a point of degree 5 and $h^0(D) \geq 2$, a nonconstant $f \in L(D)$ over $\mathbb{Q}$ has a nonzero rational pole divisor bounded by $D$; since $D$ is irreducible, it equals $D$, so $\deg f = 5$. □

Up to isogeny, $J$ is a product of modular abelian varieties. Let $\mathcal{A} \subseteq J$ be the sum of the isotypic components whose $L$-function vanishes at $s = 1$; the vanishing is decided by exact modular-symbol computations. Each isotypic factor of $J/\mathcal{A}$ has $L(1) \neq 0$, so $(J/\mathcal{A})(\mathbb{Q})$ is finite by Kato [27, Corollary 14.3] (see also Kolyvagin and Logachev [31]). By Faltings [18], the Zariski closure of $W_5^0(\mathbb{Q})$ is a finite union of translates of abelian subvarieties; a positive-dimensional component $T = x + B'$ with dense rational points is defined over $\mathbb{Q}$, and its direction $B'$ lies in $\mathcal{A}$, because the image of $T$ in $\mathrm{Pic}^5(C)/\mathcal{A}$ is a translate of an abelian variety with dense rational points in a variety with finitely many. So it suffices to show, in addition to excluding functions of degree 5, that every such $T$ has a fixed proper subdivisor on its locus with $h^0 = 1$: by Lemma 6.1 the classes of points of degree 5 then lie in the finitely many isolated points of the closure of $W_5^0(\mathbb{Q})$.

At each working prime $p \nmid N$ Magma's canonical model has good reduction (§2), and we enumerate all effective divisors of degree 5 over $\mathbb{F}_p$, for every partition of 5 into degrees of places. If $x + B \subseteq W_5^0$ over $\mathbb{Q}$, then the reduction contains the entire coset $\bar{x} + B(\mathbb{F}_p)$ in $W_5^0(\mathbb{F}_p)$.

**Lemma 6.2** (Local rigidity)**.** *Let $R = \mathbb{Z}_p$, let $Z$ be a closed subscheme of a scheme smooth over $R$, and let $U \to Z$ be a morphism from a smooth $R$-scheme of relative dimension $n$. Let $u \mapsto z$ be rational points of the special fibres. If $\dim T_z Z_{\mathbb{F}_p} = n$ and $T_u U_{\mathbb{F}_p} \to T_z Z_{\mathbb{F}_p}$ is an isomorphism, then $U \to Z$ induces an isomorphism of completed local rings at $u$ and $z$.*

*Proof.* Lifting cotangent generators gives a surjection $R[[x_1, \ldots, x_n]] \to \widehat{\mathcal{O}}_{Z,z}$; its composite with $\widehat{\mathcal{O}}_{Z,z} \to \widehat{\mathcal{O}}_{U,u} \cong R[[t_1, \ldots, t_n]]$ has invertible linear part modulo $p$, so it is an isomorphism, and the surjection is injective. □

Throughout, $W_5^r$ carries its determinantal scheme structure, defined by the same minors on the relative Picard scheme of the smooth model over $\mathbb{Z}_p$; so it is a closed subscheme of a smooth $\mathbb{Z}_p$-scheme, its special fibre is $W_5^r$ of $C_{\mathbb{F}_p}$, and tangent spaces may be computed over $\mathbb{F}_p$. At a class $[D]$ of $W_5^0$ with $h^0(D) = 1$ the annihilator of $T_{[D]}W_5^0$ in $H^0(K_C)$ is $H^0(K_C - D)$, and at a class with $h^0 \geq 2$ the tangent space of $W_5^0$ is the whole tangent space of $\mathrm{Pic}^5$. At a class $L$ of $W_5^1$ with $h^0(L) = 2$ the tangent space of $W_5^1$ is the annihilator of the image of the Petri map $H^0(L) \otimes H^0(K_C - L) \to H^0(K_C)$; if $L = A + P$ with $A$ base-point free of degree 4, the base-point-free pencil trick gives $\dim T_L W_5^1 = 8 - g + h^0(K_C - L - A)$ [3, Chapter IV, §4].

We apply Lemma 6.2 to closed subschemes of the form $Z_S = \bigcap_{s\in S}(W_5^0 - s)$ of $\mathrm{Pic}^5$ over $\mathbb{Z}_p$, where $S \subseteq J(\mathbb{Z}_p)$ is a finite set of lifts of points of $J(\mathbb{F}_p)$. If $n$ is prime to $p$, then $J[n]$ is finite étale over $\mathbb{Z}_p$, so every point of order dividing $n$ in $J(\mathbb{F}_p)$ has a unique lift to $J[n](\mathbb{Z}_p)$, and the same holds in every abelian subvariety of $J$ (these have good reduction at $p$, since their Tate modules are submodules of that of $J$). In particular, if a subgroup of $J(\mathbb{F}_p)$ of order prime to $p$ lies in the reductions of two abelian subvarieties, its lift lies in both.

6.2. **Excluding functions of degree 5.** For 21 of the 30 levels, the absence of a function of degree 5 is already in [11, §7.2 and Table 2]: at $83, 89, 92, 101, 107, 111, 131, 141, 142, 143, 167, 191$ by [11, Proposition 5.2], and at $82, 86, 99, 118, 123, 128, 145, 155, 159$ because the $\mathbb{Q}$-gonality is at least 6 [34]. It is new at $74, 77, 79, 85, 88, 103$ and at $96, 98, 100$. For completeness we give the argument we use at every level.

*The double-cover inequality.* If $C$ has an involution $w$ with quotient of genus $h$ and $g > 2h+4$, a morphism of degree 5 cannot exist, even over $\overline{\mathbb{Q}}$, by Lemma 2.2 (the degrees 2 and 5 are coprime). This applies to 17 levels (Table 2). At $142, 143, 167, 191$, [11, Proposition 5.2] argue instead with a morphism of degree 4 to $\mathbb{P}^1$: Lemma 2.2 with degrees 4 and 5 gives $g \leq 12$, while these curves have genus at least 13.

*Published gonality.* $\mathrm{gon}_{\mathbb{Q}} X_0(82) = 6$ [34, Theorem 1.4].

*Empty special loci.* At $N = 86, 99, 128$ and $p = 7, 5, 3$ respectively, the numbers of effective divisors of degree 5 over $\mathbb{F}_p$ and of their classes are equal, so $W_5^1(\mathbb{F}_p) = \varnothing$, and Lemma 2.1 applies.

*Local pencils.* At $N = 77, 85, 88$ the curve has $\mathbb{Q}$-gonality 4 [34, Theorem 1.2]; let $A$ be the class of a rational pencil of degree 4. At $p = 5, 7, 3$ respectively, the certificate checks that every effective divisor $D$ of degree 5 over $\mathbb{F}_p$ with $h^0(D) \geq 2$ has $h^0(D) = 2$, a base point, and moving part of degree 4; that these moving parts form a single class $\bar{A}_4$; and that $\dim T_{[D]}W_5^1 = 8 - g + h^0(K - D - \bar{A}_4) = 1$. The reduction $\bar{A}$ of $A$ has degree 4 and $h^0 \geq 2$ (Lemma 2.1). For a rational point $\bar{P}$, the class $\bar{A}+\bar{P}$ has $h^0 \geq 2$, so by the checks $h^0(\bar{A}+\bar{P}) = 2$ and its moving part has degree 4; hence $h^0(\bar{A}) = 2$ and $\bar{A}$ is base-point free (a base point of $\bar{A}$ would make the moving part of $\bar{A} + \bar{P}$ of degree at most 3). So $\bar{A}$ is the moving class $\bar{A}_4$, and $h^0(A) = 2$. Let $L$ be a rational class in $W_5^1$. Its reduction lies in $W_5^1(\mathbb{F}_p)$, so it is $\bar{A}_4 + \bar{P}$ for a point $\bar{P} \in C(\mathbb{F}_p)$. Apply Lemma 6.2 to $U = C \to Z = W_5^1$, $P \mapsto A + P$, at $\bar{P} \mapsto \bar{A}_4 + \bar{P}$: the Abel map is an immersion, and both tangent spaces have dimension 1. So near $\bar{A}_4 + \bar{P}$ every $\mathbb{Z}_p$-point of $W_5^1$ is $A + P$ with $P \in C(\mathbb{Z}_p)$; in particular $L = A + P$, and $P = L - A$ is a rational effective class of degree 1, that is, a rational point. Since $h^0(L) = 2 = h^0(A)$, all sections of $L$ vanish at $P$. So every rational class in $W_5^1$ has a base point, and there is no function of degree 5.

Level 74 uses the same argument at the two primes 3 and 5, after a sieve. Here $\mathrm{gon}_{\mathbb{Q}} X_0(74) = 4$ [34, Theorem 1.2]; a rational pencil of degree 4 is the pullback of the hyperelliptic pencil of $X_0(74)/w_{74}$, which has genus 2. The positive-rank part $\mathcal{A}$ is isogenous to $E_{37}^2$, where $E_{37}$ is the elliptic curve $37a1$. The rank-zero quotient $R = J/\mathcal{A}$ has $R(\mathbb{Q})$ finite, and $\#R(\mathbb{Q})$ divides

$\#R(\mathbb{F}_q)$ for every odd prime $q \neq 37$ [28, Appendix]; the gcd over the odd primes $q \leq 31$ is 513. Since $\#E_{37}(\mathbb{F}_3) = 7$ and $\#E_{37}(\mathbb{F}_5) = 8$, $\mathcal{A}(\mathbb{F}_3)$ is a 7-group and $\mathcal{A}(\mathbb{F}_5)$ a 2-group, so $\mathcal{A}(\mathbb{Q})_{\mathrm{tors}}$, which injects into both, is trivial. The differences of the four rational cusps generate a finite subgroup $\mathcal{C} \subseteq J(\mathbb{Q})$ (Manin–Drinfeld [33, 17]) whose reductions modulo 3 and 5 have order 513. It meets the torsion-free group $\mathcal{A}(\mathbb{Q})$ trivially, so it maps injectively, hence isomorphically, onto $R(\mathbb{Q})$, and $J(\mathbb{Q}) = \mathcal{C} + \mathcal{A}(\mathbb{Q})$. Every rational class of degree 5 is therefore $[5\infty] + z + a$ with $z \in \mathcal{C}$ and $a \in \mathcal{A}(\mathbb{Q})$, and its reduction lies in $[5\infty] + \bar{z} + J(\mathbb{F}_p)[\ell^\infty]$, where $\ell = 7$ for $p = 3$ and $\ell = 2$ for $p = 5$ (these primary subgroups have orders 49 and 256, and contain $\mathcal{A}(\mathbb{F}_p)$). Call $z \in \mathcal{C}$ *exceptional at* $p$ if some class in $[5\infty] + \bar{z} + J(\mathbb{F}_p)[\ell^\infty]$ lies in $W_5^1(\mathbb{F}_p)$ and either has a base-point-free pencil of degree 5 or has $\dim TW_5^1 \neq 1$. The certificate (`pencils74.m`; see also [45]) finds exactly two exceptional $z$ at each prime, and the two pairs are disjoint. For every $z$ we choose a prime at which $z$ is not exceptional; there the argument of the previous paragraph applies verbatim to every rational class $[5\infty] + z + a$ in $W_5^1$ (the certificate also checks that $W_5^1(\mathbb{F}_p)$ has a unique moving class of degree 4). So $X_0(74)$ has no function of degree 5.

The remaining levels are treated with the finiteness argument below: rank-zero lifting at $96, 98, 100$ and the quotient sieve at $79, 103$.

6.3. **Route R: direct reduction.** *Rank zero (*$96, 98, 100$*).* Here $L(J, 1) \neq 0$, so $J(\mathbb{Q})$ is finite [27, Corollary 14.3] and reduction is injective on it at the odd good primes $p = 7, 3, 3$ [28, Appendix]. This is the situation of [16, Proposition 7]. The computation shows that every special class of degree 5 over $\mathbb{F}_p$ has $h^0 = 2$ and is $\bar{A}_4 + \bar{P}$ for a class $\bar{A}_4$ of degree 4 with two sections and a point $\bar{P} \in C(\mathbb{F}_p)$, and it lifts every such $\bar{A}_4$ to a rational class $A_4$. At 96 and 100 every point of $C(\mathbb{F}_p)$ is the reduction of a rational point $P$, and injectivity forces a rational special class to be $A_4 + P$, which has the base point $P$. At 98 only four of the six points of $C(\mathbb{F}_3)$ (the cusps) lift. If $L$ is a rational special class, then $L - A_4$ is a rational class of degree 1 whose reduction is $\bar{P}$; its push-forward along the degeneracy map $X_0(98) \to X_0(49) = E$ is a point of $E(\mathbb{Q}) \cong \mathbb{Z}/2$ reducing to the image of $\bar{P}$. For a noncuspidal $\bar{P}$ that image is a noncuspidal point of $E(\mathbb{F}_3) \cong \mathbb{Z}/4$ (degeneracy maps send noncusps to noncusps), of order 4, which is not the reduction of a point of $E(\mathbb{Q})$. So $\bar{P}$ is a cusp, and $L = A_4 + P$ with a fixed rational cusp $P$.

*Empty cosets (*$85, 88, 92$*).* Here $\mathcal{A}$ is simple, of dimension $2, 1, 1$. At $p = 3$, the $\ell$-primary subgroup of $\mathcal{A}(\mathbb{F}_3)$, for $\ell = 17, 7, 7$, has no full coset inside $W_5^0(\mathbb{F}_3)$; so no translate exists.

6.4. **Route Q: a rank-zero quotient (nine prime levels).** For $N \in \{79, 83, 89, 101, 103, 107, 131, 167, 191\}$ let $Y = C/w_N$, $A = \pi^* \operatorname{Jac} Y$ and $R = J/A$; here $\mathcal{A} = A$ is simple of multiplicity one and $R$ has rank 0. Let $t = [w_N(\infty) - \infty] \in J(\mathbb{Q})$. The gcd $m$ of $\#R(\mathbb{F}_q)$ over the odd primes $q \leq 19$ bounds $\#R(\mathbb{Q})$, and at the working prime the image of $t$ in $J(\mathbb{F}_p)/A(\mathbb{F}_p)$ has order $m$ (the certificate checks $\#\langle \bar{t}, A(\mathbb{F}_p)\rangle = m \cdot \#A(\mathbb{F}_p)$). So $R(\mathbb{Q})$ is cyclic of order $m$, generated by the image of $t$, and reduction distinguishes the $m$ cosets $it + A(\mathbb{Q})$, $0 \leq i < m$. Every rational class of degree 5 lies on some $T_i = [5\infty] + it + A$, $0 \leq i < m$, and its reduction lies in $[5\infty] + i\bar{t} + A(\mathbb{F}_p)$. If $T_i \subseteq W_5^0$, then the whole coset $[5\infty] + i\bar{t} + A(\mathbb{F}_p)$ lies in $W_5^0(\mathbb{F}_p)$, and only the following indices pass this test: $i \in \{1, 2, 3, 4\}$ for $N = 79, 89, 101, 131$; $\{1, 2, 3, 4, 23\}$ for $N = 83$; $\{2, 3\}$ for $N = 103, 107, 167, 191$. For $g(Y) = 1$ these are the families $\pi^* \operatorname{Pic}^1(Y) + 3P$, $\pi^* \operatorname{Pic}^2(Y) + P$, $\pi^* \operatorname{Pic}^2(Y) + Q$, $\pi^* \operatorname{Pic}^1(Y) + 3Q$ with $P = \infty$, $Q = w_N(\infty)$, and for $g(Y) = 2$ only the middle two. The cusp is a fixed subdivisor: if $D$ contains no full fibre of $\pi$ and $d + \deg D < g - 2g(Y) + 1$, then $H^0(C, \pi^* M(D)) = H^0(Y, M)$ for every $M$ of degree $d$. Indeed, $\mathcal{O}_Y \subseteq \pi_* \mathcal{O}_C(D)$ is saturated because $D$ contains no full fibre, and the quotient is a line bundle of degree $\deg D - (g - 2g(Y) + 1)$; after tensoring with $M$ it has no sections. At $N = 83$ the index 23 is the family $\pi^* \operatorname{Pic}^1(Y) + D$ through an explicit $w_{83}$-fixed cubic point $D$: with $F = \infty + w_{83}(\infty)$ and $c = [F + D - 5\infty]$, invariance of $F + D$ under $w_{83}$, which acts as $-1$ on $R$, gives $2c = 5t$ in $R(\mathbb{Q}) \cong \mathbb{Z}/41$, so $c = 23t$;

the fixed subdivisor is $D$, which contains no full fibre of $\pi$: its three points are fixed by $w_{83}$ (checked in `cubic83.m`), so they are ramification points, whose fibres are $2Q$, and $D$ is reduced. At 79 and 103 the certificate also checks that every class of $W_5^1(\mathbb{F}_p)$ in $[5\infty] + \langle \bar{t} \rangle + A(\mathbb{F}_p)$ lies over the index 2 or 3. A rational class $L$ in $W_5^1$ reduces into this set, so it lies on $T_2$ or $T_3$; these are the families $\pi^* \operatorname{Pic}^2(Y) + P$ and $\pi^* \operatorname{Pic}^2(Y) + Q$, on which all sections vanish at the cusp $P$ or $Q$. So $L$ has a base point, and there is no function of degree 5.

6.5. **Route T: known families (fifteen levels).** For the remaining levels, $C$ has a double cover $\pi\colon C \to Y$ with $g(Y) = 2$, or a map $\pi$ of degree 4 to a curve $Y$ of genus 1 (Table 8). Let $B = \pi^* \operatorname{Jac} Y \subseteq J$ and let $F_4$ be a fibre of $\pi$ in the second case and twice a fibre in the first. Every class in $B + [F_4]$ is $\pi^*$ of a class of degree 2 or 1 on $Y$, hence effective, so

$$V = B + [F_4] + C = \{\, b + [F_4] + P : b \in B,\ P \in C \,\} \subseteq W_5^0,$$

the image of the proper map $B \times C \to \operatorname{Pic}^5(C)$; it is closed, of dimension $\dim B + 1$, and stable under translation by $B$.

*The certificate* (see also [45]). At the prime $p$ of Table 8, a set of subgroups $S \subseteq J(\mathbb{F}_p)$ is tested: for a list of primes $\ell \neq p$ (and $n = 8$ at $N = 86$), all subgroups of $J(\mathbb{F}_p)$ of order $\ell$, chosen so that every simple factor $A_i$ of $\mathcal{A}$ has $\#A_i(\mathbb{F}_p)$ divisible by some tested $\ell$. For every tested $S$ and every full coset $x + S \subseteq W_5^0(\mathbb{F}_p)$ the certificate checks

(i) $S \subseteq B(\mathbb{F}_p)$;
(ii) $x + S \subseteq B(\mathbb{F}_p) + [F_4] + C(\mathbb{F}_p)$;
(iii) $\dim T_x Z_S = \dim B + 1$, where $T_x Z_S$ is computed as the common annihilator of the spaces $H^0(K_C - D)$ for the classes $x + s = [D]$, $s \in S$, with $h^0(D) = 1$;
(iv) the differential of $B \times C \to \operatorname{Pic}^5(C)$ is injective at every point of $B(\mathbb{F}_p) \times C(\mathbb{F}_p)$.

At 82, 99 and 128 the positive-rank part $\mathcal{A}$ is a single elliptic curve of multiplicity one, which is $B$ (identified through its newform); there the only test is $S = B(\mathbb{F}_p)$.

*Every translate lies in $V$.* Let $T = t + B' \subseteq W_5^0$ be a positive-dimensional translate with $T(\mathbb{Q})$ dense, as in the framework, with $t \in T(\mathbb{Q})$; then $B' \subseteq \mathcal{A}$. Suppose first that we are not at $82, 99, 128$. The abelian variety $B'$ contains a simple abelian subvariety isogenous to some $A_i$, so $\#B'(\mathbb{F}_p)$ is divisible by a tested prime $\ell$, and $B'(\mathbb{F}_p)$ contains a subgroup $S$ of order $\ell$, which is tested. (At 86, $\mathcal{A}$ is isogenous to $E_{43}^2$ with $E_{43} = 43a1$, the only curve in its isogeny class, so $B'$ contains a copy of $E_{43}$, and $E_{43}(\mathbb{F}_7) \cong \mathbb{Z}/8$ gives a cyclic subgroup of order 8.) The reduction of $T$ is the translate $\bar{t} + B'_{\mathbb{F}_p}$, which lies in $(W_5^0)_{\mathbb{F}_p}$, so $\bar{t} + S$ is a full coset in $W_5^0(\mathbb{F}_p)$ and (i)–(iv) hold for it. Let $\tilde{S} \subseteq J(\mathbb{Z}_p)$ be the lift of $S$; as $\ell \neq p$, it lies in both $B'$ and $B$ (by (i)). Hence $T + s = T$ and $V + s = V$ for $s \in \tilde{S}$, so $T$ and $V$ both lie in $Z_{\tilde{S}} = \bigcap_{s \in \tilde{S}} (W_5^0 - s)$. At $82, 99, 128$ we have $B' = B$; we lift the points of $S = B(\mathbb{F}_p)$ to arbitrary points of $B(\mathbb{Z}_p)$ (possible since $B$ is smooth), and again $T, V \subseteq Z_{\tilde{S}}$, because both are stable under all of $B$.

By (ii), $\bar{t} = \bar{b} + [F_4] + \bar{P}$ with $\bar{b} \in B(\mathbb{F}_p)$ and $\bar{P} \in C(\mathbb{F}_p)$; lift $(\bar{b}, \bar{P})$ to a point $u$ of $B \times C$ over $\mathbb{Z}_p$. Apply Lemma 6.2 to $B \times C \to Z_{\tilde{S}}$ at $\bar{u} \mapsto \bar{t}$: by (iii) and (iv) the tangent map is an isomorphism onto a space of dimension $\dim B + 1$. So the completed local ring of $Z_{\tilde{S}}$ at $\bar{t}$ is that of $B \times C$, and every $\mathbb{Z}_p$-point of $Z_{\tilde{S}}$ reducing to $\bar{t}$ lies in $V$. The points of $T(\mathbb{Q}_p) = T(\mathbb{Z}_p)$ reducing to $\bar{t}$ form a nonempty $p$-adic open subset of the smooth irreducible variety $T$, which is Zariski dense in $T$; as $V$ is closed, $T \subseteq V$.

*No translate contributes points of degree 5.* Let $q\colon \operatorname{Pic}^5(C) \to \operatorname{Pic}^5(C)/B$ be the quotient, a torsor under $J/B$. Then $q(V)$ is the image of the curve $C$, whose differences generate $J/B$, and $q(T)$ is a translate of the abelian variety $q(B')$. In every row of Table 8, $\dim(J/B) = g - \dim B > 1$. If $q(B') \neq 0$, then $q(T)$ would be a curve, hence equal to the irreducible curve $q(V)$, and the differences of points of $q(V)$ would generate only the one-dimensional $q(B')$; so $q(T)$ is a point.

Hence $T \subseteq B + [F_4] + P$ for a single point $P \in C(\overline{\mathbb{Q}})$ ($T$ is irreducible and the fibres of $C \to q(V)$ are finite). Since $T$ is defined over $\mathbb{Q}$, also $T \subseteq B + [F_4] + \sigma P$ for every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. By the exclusion of functions of degree 5 and Lemma 6.1, a class of a point of degree 5 has $h^0 = 1$. If $T \subseteq W_5^1$, then $T$ contains no such class. Otherwise $T \setminus W_5^1$ is a nonempty open subset of $T$, so its rational points are dense in $T$. Let $L \in T(\mathbb{Q})$ with $h^0(L) = 1$, and let $D_L$ be its unique effective divisor. As $L$ has an effective representative of the form $\pi^*(e) + \sigma P$ for each $\sigma$, $D_L$ contains the Galois orbit $O$ of $P$, so $\deg O \leq 5$. If $\deg O = 5$, then $D_L = O$ for all such $L$; these $L$ are dense in $T$, so $T$ would be a point. Hence $O$ is a rational effective divisor of degree at most 4 contained in every $D_L$, and no $D_L$ is a point of degree 5.

*Proof of Theorem A.* Combine the function exclusions with the three routes, as in Table 2. The 36 certificates are in the accompanying repository (`certificates/A_x0_quintic`), with a list of the data per level; the supplementary write-up [45] gives the computation at each level in more detail. □

TABLE 2. Theorem A: the thirty levels. The route proves finiteness of the points of degree 5 assuming there is no function of degree 5; the last column is the separate argument excluding functions of degree 5 ("double cover": the inequality $g > 2h + 4$).

| $N$ | $g$ | route | degree-5 functions | $N$ | $g$ | route | degree-5 functions |
|---|---|---|---|---|---|---|---|
| 74 | 8 | T | local pencil, 2 primes | 107 | 9 | Q | double cover |
| 77 | 7 | T | local pencil | 111 | 11 | T | double cover |
| 79 | 6 | Q | quotient sieve | 118 | 14 | T | double cover |
| 82 | 9 | T | [34] | 123 | 13 | T | double cover |
| 83 | 7 | Q | double cover | 128 | 9 | T | $W_5^1(\mathbb{F}_3) = \varnothing$ |
| 85 | 7 | R | local pencil | 131 | 11 | Q | double cover |
| 86 | 10 | T | $W_5^1(\mathbb{F}_7) = \varnothing$ | 141 | 15 | T | double cover |
| 88 | 9 | R | local pencil | 142 | 17 | T | double cover |
| 89 | 7 | Q | double cover | 143 | 13 | T | double cover |
| 92 | 10 | R | double cover | 145 | 13 | T | double cover |
| 96 | 9 | R | rank zero | 155 | 15 | T | double cover |
| 98 | 7 | R | rank zero | 159 | 17 | T | double cover |
| 99 | 9 | T | $W_5^1(\mathbb{F}_5) = \varnothing$ | 167 | 14 | Q | double cover |
| 100 | 7 | R | rank zero | 191 | 16 | Q | double cover |
| 101 | 8 | Q | double cover | | | | |
| 103 | 8 | Q | quotient sieve | | | | |

## 7. Points of degree 5 on intermediate curves: proof of Theorem B

### 7.1. **Reduction to finitely many curves.**

**Lemma 7.1** (Projection)**.** *Let $C \to D$ be a finite morphism over $\mathbb{Q}$. If $D$ has only finitely many points of degree 1 and of degree 5, then $C$ has only finitely many points of degree 5.*

*Proof.* The image of a point of degree 5 has degree dividing 5, and fibres are finite. □

Forgetful maps $X_\Delta(N) \to X_{\Delta'}(M)$, for $M \mid N$ and $\Delta$ mapping into $\Delta'$, are defined over $\mathbb{Q}$; in particular every $X_\Delta(N)$ maps to $X_0(N)$. By Theorem A and [11, Theorem 1.10], $X_0(N)$ has finitely many points of degree 5 unless

$$N \in \{1, \dots, 45\} \cup \{47, \dots, 58\} \cup \{61, 63, 64, 65, 67, 68, 72, 73, 75, 80, 81, 91, 109, 121, 125\},$$

and every $X_0(N)$ with $N$ outside this list has genus $> 1$ (otherwise its rational cusp would give a function of degree 5, §3.1), hence finitely many rational points by Faltings [18]; so these $X_0(N)$ can serve as targets in Lemma 7.1. So every strict pair with $N \geq 192$, and 807 of the 1011 strict pairs with $N \leq 191$, have finitely many points of degree 5. The strict pairs with $N \leq 191$ are enumerated twice, once from primitive rows $(c, d)$ modulo $N$ up to $\Delta$ and once with Sage's `GammaH`; full sets of elements, not orders or genera, are used to identify them. Seven more pairs map to intermediate curves already shown to be negative (Table 6), and fifteen more, of genus 57 to 256, satisfy $119\mu > 120000$, so they have $\mathrm{gon}_{\mathbb{C}} > 10$ and finitely many points of degree 5 by Lemmas 2.5 and 2.6 and [14, Proposition 2.2] (code S in Table 5). This leaves 182 strict pairs.

7.2. **The 43 positive curves.** Each curve in Table 4 has a function of degree 5 over $\mathbb{Q}$, hence infinitely many points of degree 5 (Lemma 2.4):

- L (13 curves, $g \leq 2$): for a rational cusp $P$, $\ell(5P) = 6 - g > \ell(4P)$;
- K (14 curves, $g = 3, 4$): in genus 3, 5 is not a Weierstrass gap at the cusp $\infty$ (read off from the orders of vanishing of a basis of cusp forms); in genus 4 the curve is not hyperelliptic (the multiplication map $\mathrm{Sym}^2 H^0(K) \to H^0(2K)$ has rank $3g - 3 = 9$, checked on $q$-expansions up to the Sturm bound). Then for a rational cusp $P$ the net $|K - P|$ of degree 5 is base-point free (a base point $Q$ would give $h^0(P + Q) = 2$), so it maps $C$ birationally onto a plane quintic, and projecting from a point of $\mathbb{P}^2(\mathbb{Q})$ not on this quintic gives a morphism of degree 5 to $\mathbb{P}^1$;
- W (12 curves): an explicit divisor $D$ of degree 5 with $\ell(D) = 2$ and no base point, on a canonical model certified through the Sturm bound and a Hilbert polynomial comparison;
- P (4 curves): the $\mathbb{Q}$-pentagonal curves of [37, Theorems 1.4–1.5].

7.3. **The negative curves.** For the remaining 139 curves we show finiteness. The arguments are listed by code in Table 5, which also lists the 15 curves of large index (code S) treated above.

**Lemma 7.2** (Rank zero)**.** *If $C(\mathbb{Q}) \neq \varnothing$, $J(C)(\mathbb{Q})$ is finite and $C$ has no function of degree 5 over $\mathbb{Q}$, then $C$ has finitely many points of degree 5.*

*Proof.* As in Lemma 6.1, each point of degree 5 is the unique effective divisor in its class, and there are finitely many classes. □

*Rank zero* (82 curves). We show $J(\mathbb{Q})$ finite from nonvanishing $L$-values (exact modular-symbol $L$-ratios), using Kato [27, Corollary 14.3] as applied to modular Jacobians with nontrivial character in [14, §4]. For 25 of the 74 curves of class Z we use $L(J_1(N), 1) \neq 0$: the forgetful map $X_1(N) \to X_\Delta(N)$ makes $J_\Delta(N)$ an isogeny quotient of $J_1(N)$, so $J_\Delta(N)(\mathbb{Q})$ is finite as well. The absence of a function of degree 5 comes from

- Z (74 curves) Orlić's classification: genus $\geq 5$ and not in Orlić's lists of curves of $\mathbb{Q}$-gonality at most 5, so $\mathrm{gon}_{\mathbb{Q}} \geq 6$ [37];
- B (2 curves, $35\langle -1, 8\rangle$ and $55\langle -1, 4\rangle$, of genus 7 and 9) the curves are bielliptic [25, Theorem 0.2], so a function of degree 5 would violate Lemma 2.2: $g \leq 2 + 4 = 6$;
- T (3 curves) Petri lifting: over $\mathbb{F}_p$ every class of $W_5^1$ is $A + P$ for an explicit rational base-point free class $A$ of degree 4, with Petri tangent dimension 1; Lemma 6.2 identifies $W_5^1$ near each such class with the Abel curve $A + C$, so a rational pencil of degree 5 has a base point;
- R (3 curves; see also [44]) a complete sieve of rational divisor classes: an explicit subgroup of $J(\mathbb{Q})$ generated by differences of rational points is shown to be all of $J(\mathbb{Q})$, and every class of $W_5^1(\mathbb{F}_p)$ in its reduction is lifted and checked over $\mathbb{Q}$ (each turns out to be a

rational class $D_4 + P$ with a base point $P$). At level 40 ($\Delta = \langle -1, 3\rangle$, genus 7) the explicit subgroup is generated by the differences of six rational points and has order 4032. An upper bound for $\#J(\mathbb{Q})$ comes from the Hecke method of [10, §4.3]. Rational torsion is fixed by complex conjugation, and for a prime $\ell \nmid 40$ the Eichler–Shimura relation shows that $1 - T_\ell + \ell\langle\ell\rangle = 1 - T_\ell + T_\ell^2 - T_{\ell^2}$ kills rational torsion of order prime to $\ell$. The operators $c - 1$, with $c$ complex conjugation, and these Eichler–Shimura operators act on $H_1(X, \mathbb{Z})$, computed from Manin symbols, and the rational torsion of order prime to the primes used embeds into the common kernel of these operators on $H_1(X, \mathbb{Q})/H_1(X, \mathbb{Z}) \cong (\mathbb{Q}/\mathbb{Z})^{2g}$, a finite group computed by a Smith normal form. The relation at $\ell$ says nothing about $\ell$-primary torsion, so we compute five such kernels: one with all four primes $\ell \in \{3, 7, 11, 13\}$, which bounds the primary parts for primes other than these four, and for each of the four primes $\ell$ one with the other three primes, which bounds the $\ell$-primary part. All five kernels are isomorphic to $\mathbb{Z}/2 \times \mathbb{Z}/24 \times \mathbb{Z}/168$, of order $8064 = 2^7 \cdot 3^2 \cdot 7$, so $\#J(\mathbb{Q})$ divides 8064. Reduction is injective on $J(\mathbb{Q})$ at odd good primes, and the $\ell$-parts of $\#J(\mathbb{F}_p)$ for $p \le 31$, $p \nmid 40\ell$, give $\#J(\mathbb{Q}) \mid 2^6 \cdot 3^2 \cdot 7^2 = 28224$. Hence $\#J(\mathbb{Q})$ divides $\gcd(8064, 28224) = 4032$, and $J(\mathbb{Q})$ is the explicit subgroup.

At level 48 the two curves ($\Delta = \langle -1, 5\rangle$ and $\Delta = \langle -1, 7, 17\rangle$, both of genus 7) have explicit subgroups generated by cuspidal divisors, of orders 2304 and 12800, and these are also the upper bounds. Let $\pi\colon X_1(48) \to X = X_\Delta(48)$ be the forgetful map, of degree 4. The pullback $\pi^*\colon J(\mathbb{Q}) \to J_1(48)(\mathbb{Q})$ is induced on homology by the topological transfer $H_1(X, \mathbb{Z}) \to H_1(X_1(48), \mathbb{Z})$, the sum of the four lifts of a cycle, which we compute on Manin symbols (and check that it is integral, that $\pi_* \circ \pi^* = 4$, and that it commutes with complex conjugation and with $T_7$, $T_{11}$). By [10, Theorem 4.13], $J_1(48)(\mathbb{Q})_{\mathrm{tors}}$ is the rational cuspidal group, which we compute in the same Manin-symbol model. Point counts give an integer $b$ ($2^{13} \cdot 3^4$ and $2^{13} \cdot 5^2$) with $b\,J(\mathbb{Q}) = 0$, and the primes $\ell \in \{7, 11, 13, 17, 19\}$ are prime to $b$, so the Eichler–Shimura relations at these primes kill all of $J(\mathbb{Q})$; the common kernel of $c - 1$, multiplication by $b$ and these relations on $H_1(X, \mathbb{Q})/H_1(X, \mathbb{Z})$ has order 36864 and 204800 respectively. Every rational torsion class of $J$ lies in the subgroup of this kernel whose transfer lies in the rational cuspidal group of $J_1(48)$. This inverse image has order 2304 for $\Delta = \langle -1, 5\rangle$ and 12800 for $\Delta = \langle -1, 7, 17\rangle$. The argument does not need $\pi^*$ to be injective.

*Positive rank* (57 curves). None of these curves has a function of degree 5 over $\mathbb{Q}$: they have genus at least 8 and are not in the lists of curves of $\mathbb{Q}$-gonality at most 5 [24], [37, Theorems 1.3–1.5], so their $\mathbb{Q}$-gonality is at least 6. The five curves of class F and the curve of class C are treated directly below. For the other 51 curves (classes E, E0 and D) we use the classification of curves with infinitely many points of degree at most 4 [13, Theorems 1.4–1.6]: none of these 51 curves has infinitely many points of degree $\le 4$, so if one of them had infinitely many points of degree 5, its minimum density degree $\min \delta(X/\mathbb{Q})$ would be 5. We use two results of Kadets and Vogt for a nice curve $X$ of genus $g$ over a number field $k$.

- [26, Theorem 1.3]: *if* $\min \delta(X/k) = d$, $m = \lceil d/2 \rceil - 1$ *and* $\varepsilon = 3d - 1 - 6m$, *then either there is a nonconstant morphism* $\varphi\colon X \to Y$ *of degree at least* 2 *with* $d = \min \delta(Y/k) \cdot \deg \varphi$, *or* $g \le \max\bigl(\frac{d(d-1)}{2} + 1,\ 3m(m-1) + m\varepsilon\bigr)$.
- [26, Proposition 6.3 and Table 1]: *if* $\min \delta(X/k) = 5$, *then* $X$ *is a cover of degree* 5 *of* $\mathbb{P}^1$ *or of an elliptic curve of positive rank, or the normalization of a Debarre–Fahlaoui curve, or* $g \in \{5, 6, 7, 8\}$. Here a Debarre–Fahlaoui curve [26, Definition 5.1], [7] is a geometrically integral curve on $\mathrm{Sym}^2 E$, for an elliptic curve $E/k$ of positive rank, in the numerical class $(d + j)H - jF$ with $1 \le j \le d$, where $H$ is the class of the divisors $\{x + y : x = x_0\}$ and $F$ the class of the fibres of the addition map $\mathrm{Sym}^2 E \to E$.

For $d = 5$ we have $m = 2$ and $\varepsilon = 2$, so the bound in Theorem 1.3 is $g \le \max(11, 10) = 11$; and since 5 is prime, a morphism as in its first alternative has degree 5 to a curve $Y$ with $\min \delta(Y/\mathbb{Q}) = 1$, that is, to $\mathbb{P}^1$ or to an elliptic curve of positive rank ([13, Corollary 2.5] states this for $g \ge 12$). So for $g \ge 12$ it suffices to exclude these two kinds of maps. For $9 \le g \le 11$ Proposition 6.3 adds only Debarre–Fahlaoui curves. On $\mathrm{Sym}^2 E$ we have $H^2 = H \cdot F = 1$, $F^2 = 0$ and $K = -2H + F$, so a curve of class $(5+j)H - jF$ has arithmetic genus $11 - j(j-1)/2$ by adjunction, and $F \cdot ((5+j)H - jF) = 5 + j$. Its normalization has genus at most $11 - j(j-1)/2$, so $g \ge 9$ forces $j \le 2$, and the addition map gives a morphism $X \to E$ of degree 6 or 7 to an elliptic curve of positive rank. The curves of genus 8 ($53\langle -1, 4\rangle$ and $61\langle -1, 4\rangle$) are in class F and do not use this argument.

E (43 curves, genus $\ge 12$, so only elliptic maps of degree 5 need to be excluded) every positive-rank elliptic factor $E$ has conductor $N$, multiplicity one and odd analytic rank (all elliptic curves of conductor less than 389 have analytic rank 0 or 1, and those of analytic rank 0 have rank 0). For such $E$ and $N \le 800$, the composition $X_\Delta(N) \to X_0(N) \to E$ with the optimal parametrization of $X_0(N)$ is a strong Weil parametrization of $X_\Delta(N)$, of degree $t\, d_E$, where $t = [(\mathbb{Z}/N\mathbb{Z})^\times : \Delta]$ and $d_E$ is the modular degree [13, Corollary 4.10]. Every morphism from $X_\Delta(N)$ to an elliptic curve $E'$ isogenous to $E$ factors, up to translation, through this strong Weil parametrization, as in the proof of [13, Corollary 4.12], so its degree is divisible by $t\, d_E$. In all these cases $t, d_E > 1$ (Table 10), so $t\, d_E$ is composite and the degree is not 5. At levels 67 and 73, which are prime, the elliptic curves of conductor $N$ have nonzero central $L$-value, hence rank 0; so these seven curves have no positive-rank elliptic quotient at all and the condition is vacuous. They could equally be listed under E0; we keep the codes of the first version;

E0 (5 curves of genus 16 to 64, at levels 109 and 125) the decomposition of the Jacobian has no positive-rank elliptic factor, which is certified by a separate computation of the elliptic factors;

D (3 curves of genus 9 or 10: $37\langle -1, 10\rangle$, $57\langle -1, 2\rangle$, $73\langle -1, 6\rangle$) here a Debarre–Fahlaoui configuration must also be excluded, that is, elliptic maps of degree 5, 6 or 7: at level 73 there is no positive-rank elliptic factor, and at levels 37 and 57 every elliptic map of positive rank has degree at least 12 and 8 respectively [13, Corollary 4.10];

C (1 curve, $43\langle -1, 2\rangle$) every simple factor of the Jacobian other than one copy of the elliptic curve $A = 43a1$ has nonzero central $L$-value, so a positive-dimensional translate in $W_5^0$ with dense rational points has the form $x + A$ with $x$ rational; its reduction modulo 2 contains the full coset $\bar{x} + A(\mathbb{F}_2)$, of order 5, in $W_5^0(\mathbb{F}_2)$. The set $W_5^0(\mathbb{F}_2)$ has 2133 elements, and $J(\mathbb{F}_2) \cong \mathbb{Z}/38 \times \mathbb{Z}/1330$ has a unique subgroup of order 5, which is therefore $A(\mathbb{F}_2)$; no coset of it lies in $W_5^0(\mathbb{F}_2)$. So $W_5^0(\mathbb{Q})$ is finite;

F (5 curves, levels 53, 61, 65) there is a map $f\colon C \to E$ of degree 4 whose pullback is the positive-rank part $A$, and the surface $Z = \{f^*Q + P\}$ of reducible divisors lies in $W_5^0$. At a good odd prime $p$, with $B = A(\mathbb{F}_p)$, we check that every full $B$-coset in $W_5^0(\mathbb{F}_p)$ lies in $Z$, that the intersection of the corresponding tangent spaces has dimension at most 2, and that $A \times C \to \mathrm{Pic}^5 C$ has injective differential at the relevant points. Since $f_* f^* = [4]$ and $p$ is odd, $A$ has a smooth model over $\mathbb{Z}_p$. We lift each point of $B$ to some point of $A(\mathbb{Z}_p)$; for the two curves of genus 11 at level 65 we use $p = 3$, which divides $\#A(\mathbb{F}_3) = 6$, so these lifts need not be torsion or form a subgroup, but nothing requires this: a rational translate of $A$ in $W_5^0$ and the surface $Z$ are both stable under all of $A$, so both lie in $\bigcap_b (W_5^0 - a_b)$, and Lemma 6.2 puts every rational translate of $A$ in $W_5^0$ into $Z$. A point of degree 5 in $Z$ would be an effective divisor $f^*Q + P$; two distinct fibres of $f$ are disjoint and together have degree $8 > 5$, so the fibre $f^*Q$ and the point $P$ are

unique, hence Galois-invariant, and $P$ would be a rational point in the support, which is impossible.

*Proof of Theorem B.* The 43 positive curves, the 139 curves above, the 7 further projections, the 15 curves with large index and the 807 projections to $X_0(N)$ account for all 1011 strict pairs with $N \leq 191$, and projection to $X_0(N)$ handles $N \geq 192$. □

*Proof of Corollary E.* The 39 positive curves other than Orlić's four $\mathbb{Q}$-pentagonal curves have $\mathbb{Q}$-gonality at most 4: those of genus $g \leq 4$ have a rational cusp, so $\mathrm{gon}_{\mathbb{Q}} \leq \max(g, 2) \leq 4$ (§3.1), and the twelve curves of type W are $\mathbb{Q}$-tetragonal [37, Theorem 1.3]. By Lemma 2.4 they have minimum density degree at most 4. The four pentagonal curves have genus 6, 6, 8 and 16, so finitely many rational points by Faltings [18], and none of them occurs in the lists of [13, Theorems 1.4, 1.5 and 1.6] of curves with infinitely many points of degree 2, 3 or 4. By Theorem B they have infinitely many points of degree 5. □

Every case has a certificate or a precise citation in the accompanying repository (`certificates/B_intermediate_quintic`), which also contains the full case list; the supplementary write-up [44] describes the certificates of the special cases (classes T, R, C, F and W) in more detail.

## 8. Computations

All computations use Magma V2.29-4 [5], SageMath 10.7 with MD Sage [9] (commit `6022429`), CurveArith [15] (commit `b7c6192`) and Zywina's package [46] (commit `ce31d78`); the three packages are pinned as git submodules of the accompanying repository. The repository is organized by theorem:

| folder | contents |
|---|---|
| `data/` | one row per curve: bounds, methods, certificates or citations |
| `certificates/A_x0_quintic/` | Theorem A: 36 Magma certificates and logs |
| `certificates/B_intermediate_quintic/` | Theorem B and Corollary E: all 1388 pairs $(N, \Delta)$, $N \leq 191$ (1011 strict) |
| `certificates/C_intermediate_gonality/` | Theorem C: models, CurveArith runs, upper-bound certificates |
| `certificates/C_43.132.9.a.2/` | Proposition 4.1 |
| `certificates/D_x1_gonality/` | Theorem D: Phase 1 driver, Phase 2 program, logs |
| `certificates/good_reduction.m` | good reduction of all canonical models used (§2) |
| `verify.py` | checks that every row of `data/` has a passing log or a citation |

**Method II for $X_1(N)$.** Phase 1 is a short Magma script that loads CurveArith and calls its internal function `PrecomputePowerseriesExpansions` with exactly the places and precisions that `CAHasFunctionOfDegreeAtMost` would use; no CurveArith code is modified. It runs in several Magma processes. Every process enumerates all places, as CurveArith does, and computes the expansions only at the places assigned to it: the place's key (the irreducible polynomial $g(t) \in \mathbb{F}_p[t]$ below it, encoded as an integer, or 0 at infinity) is hashed by a fixed integer function, and the hash determines the process. This assignment does not depend on the order in which Magma lists the places, so the processes together treat every place exactly once; the merge checks that the counts add up, and that the processes agree on the curve, the place counts and the basis of differentials. (Splitting by position in the list instead is not safe: Magma does not always list the places in the same order in different processes.) Its output is the list of matrices of Lemma 3.1.

Phase 2 is a C++ program of about 1000 lines (with GMP) that enumerates $\mathcal{F}_b(d)$, performs the incremental Gaussian elimination of §3.3 over $\mathbb{F}_p$, and enumerates the kernel vectors up to scalars (Lemma 3.4). It is a reimplementation, for speed, of the option `Al:="Cover"` of our patch to CurveArith (§3.3), which we used to cross-check it on small curves. The baseline $b$ is chosen to minimize the number of divisors times the size of the kernels. For any $b$ the result is correct; it only affects the running time.

**Validation.** The pipeline was tested in both directions against CurveArith's unmodified `CAGonality`: on $X_1(26)$ over $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$ and on $X_1(29)$ over $\mathbb{F}_2, \mathbb{F}_3$, it finds no function of degree $\mathrm{gon}_{\mathbb{F}_p} - 1$ and finds one of degree $\mathrm{gon}_{\mathbb{F}_p}$. When a function is found, its divisor is checked independently by a rank computation. Each Phase 1 process also computes a few Riemann–Roch dimensions with Magma's `Dimension`, for divisors supported on its own places; Phase 2 checks them against Lemma 3.1 before searching.

**Running times.** The 36 certificates of Theorem A take about 40 minutes in total, the longest ($N = 86$) 16 minutes; these are wall-clock times of each script, measured with eight scripts running in parallel on a shared machine, and an independent rerun took about half as long. The certificates of Proposition 4.1 take a few minutes. The lower bounds of Theorem C take about 2 hours in total, dominated by 48.384.19.*bc*.1 (66 minutes). The running times for Theorem D are in Table 1.

**Use of AI tools.** Large language model assistants (OpenAI's GPT-6 Astra and GPT-6.1 SOL, through Codex, and Anthropic's Claude, through Claude Code) were used to write and optimize code, to organize the certificates, to draft parts of the text, and to review drafts of the revision. The proofs are built to be checked without trusting these tools: each step is a cited published result, a short argument written out in full, or a computation backed by a certificate in the accompanying repository that can be rerun independently. The author takes responsibility for the content of this paper.

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## Appendix A. Tables

Table 3: $\mathbb{Q}$-gonality of the strict curves $X_\Delta(N)$, $N \leq 50$. Lower and upper bound: method (§3), or "genus" for $g \leq 1$ or the low-genus arguments of §3.1. For $N \leq 40$ and $g \geq 3$ both bounds are from [37, Theorem 1.2, Table 1].

| label | $\Delta$ | $g$ | $\mathrm{gon}_{\mathbb{Q}}$ | lower bound | upper bound |
|---|---|---|---|---|---|
| 13.28.0.a.1 | $\langle -1, 3\rangle$ | 0 | 1 | genus | genus |
| 13.42.0.a.2 | $\langle -1, 5\rangle$ | 0 | 1 | genus | genus |
| 15.48.1.a.1 | $\langle -1, 4\rangle$ | 1 | 2 | genus | genus |
| 16.48.0.v.1 | $\langle -1, 7\rangle$ | 0 | 1 | genus | genus |
| 17.36.1.a.2 | $\langle -1, 2\rangle$ | 1 | 2 | genus | genus |
| 17.72.1.a.2 | $\langle -1, 4\rangle$ | 1 | 2 | genus | genus |
| 19.60.1.a.2 | $\langle -1, 7\rangle$ | 1 | 2 | genus | genus |
| 20.72.1.f.2 | $\langle -1, 9\rangle$ | 1 | 2 | genus | genus |
| 21.64.1.a.3 | $\langle -1, 4\rangle$ | 1 | 2 | genus | genus |
| 21.96.3.a.1 | $\langle -1, 8\rangle$ | 3 | 2 | [37] | [37] |
| 24.96.1.dg.2 | $\langle -1, 11\rangle$ | 1 | 2 | genus | genus |
| 24.96.3.ge.2 | $\langle -1, 5\rangle$ | 3 | 3 | [37] | [37] |
| 24.96.3.gi.1 | $\langle -1, 7\rangle$ | 3 | 3 | [37] | [37] |
| 25.60.0.a.1 | $\langle -1, 4\rangle$ | 0 | 1 | genus | genus |
| 25.150.4.f.1 | $\langle -1, 7\rangle$ | 4 | 4 | [37] | [37] |
| 26.84.4.a.1 | $\langle -1, 3\rangle$ | 4 | 3 | [37] | [37] |
| 26.126.4.a.1 | $\langle -1, 5\rangle$ | 4 | 3 | [37] | [37] |
| 27.108.1.a.1 | $\langle -1, 8\rangle$ | 1 | 2 | genus | genus |
| 28.96.4.c.4 | $\langle -1, 3\rangle$ | 4 | 3 | [37] | [37] |
| 28.144.4.d.1 | $\langle -1, 13\rangle$ | 4 | 3 | [37] | [37] |
| 29.60.4.a.1 | $\langle -1, 4\rangle$ | 4 | 3 | [37] | [37] |
| 29.210.8.b.2 | $\langle -1, 12\rangle$ | 8 | 6 | [37] | [37] |
| 30.144.5.a.1 | $\langle -1, 11\rangle$ | 5 | 4 | [37] | [37] |
| 31.96.6.a.1 | $\langle -1, 2\rangle$ | 6 | 5 | [37] | [37] |
| 31.160.6.c.1 | $\langle -1, 5\rangle$ | 6 | 5 | [37] | [37] |
| 32.96.1.f.1 | $\langle -1, 7\rangle$ | 1 | 2 | genus | genus |
| 32.192.5.bu.1 | $\langle -1, 15\rangle$ | 5 | 4 | [37] | [37] |
| 33.96.5.a.4 | $\langle -1, 2\rangle$ | 5 | 4 | [37] | [37] |
| 33.240.11.a.1 | $\langle -1, 10\rangle$ | 11 | 6 | [37] | [37] |
| 34.108.5.a.2 | $\langle -1, 9\rangle$ | 5 | 4 | [37] | [37] |
| 34.216.9.a.1 | $\langle -1, 13\rangle$ | 9 | 6 | [37] | [37] |
| 35.96.5.a.2 | $\langle -1, 4\rangle$ | 5 | 4 | [37] | [37] |
| 35.144.7.a.2 | $\langle -1, 8\rangle$ | 7 | 4 | [37] | [37] |
| 35.192.9.b.3 | $\langle -1, 11\rangle$ | 9 | 6 | [37] | [37] |
| 35.288.13.a.2 | $\langle -1, 6\rangle$ | 13 | 8 | [37] | [37] |
| 36.144.3.c.1 | $\langle -1, 11\rangle$ | 3 | 3 | [37] | [37] |
| 36.216.7.u.1 | $\langle -1, 17\rangle$ | 7 | 4 | [37] | [37] |
| 37.76.4.a.2 | $\langle -1, 3\rangle$ | 4 | 3 | [37] | [37] |

| label | $\Delta$ | $g$ | $\mathrm{gon}_{\mathbb{Q}}$ | lower bound | upper bound |
|---|---|---|---|---|---|
| 37.114.4.b.1 | $\langle -1, 8\rangle$ | 4 | 3 | [37] | [37] |
| 37.228.10.a.2 | $\langle -1, 10\rangle$ | 10 | 6 | [37] | [37] |
| 37.342.16.c.2 | $\langle -1, 6\rangle$ | 16 | 9 | [37] | [37] |
| 38.180.10.a.1 | $\langle -1, 7\rangle$ | 10 | 6 | [37] | [37] |
| 39.112.5.a.2 | $\langle -1, 4\rangle$ | 5 | 4 | [37] | [37] |
| 39.168.9.a.1 | $\langle -1, 5\rangle$ | 9 | 4 | [37] | [37] |
| 39.224.9.a.3 | $\langle -1, 16\rangle$ | 9 | 6 | [37] | [37] |
| 39.336.17.c.1 | $\langle -1, 14\rangle$ | 17 | 8 | [37] | [37] |
| 40.144.5.gn.1 | $\langle -1, 9, 11\rangle$ | 5 | 4 | [37] | [37] |
| 40.144.7.fp.1 | $\langle -1, 3\rangle$ | 7 | 4 | [37] | [37] |
| 40.144.7.fs.1 | $\langle -1, 7\rangle$ | 7 | 4 | [37] | [37] |
| 40.288.13.sp.1 | $\langle -1, 9\rangle$ | 13 | 8 | [37] | [37] |
| 40.288.13.sq.2 | $\langle -1, 11\rangle$ | 13 | 7 | [37] | [37] |
| 40.288.9.bh.1 | $\langle -1, 19\rangle$ | 9 | 6 | [37] | [37] |
| 41.84.5.a.1 | $\langle -1, 2\rangle$ | 5 | 4 | II, $p = 3$ | I (pullback) |
| 41.168.11.b.1 | $\langle -1, 4\rangle$ | 11 | 6 | II, $p = 2$ | I (cusps) |
| 41.210.11.b.1 | $\langle -1, 3\rangle$ | 11 | 8 | II, $p = 5$ | I (cusps) |
| 41.420.21.d.1 | $\langle -1, 9\rangle$ | 21 | 11 | II, $p = 2$ | I (cusps) |
| 42.192.9.c.2 | $\langle -1, 5\rangle$ | 9 | 6 | II, $p = 5$ | I (pullback) |
| 42.288.13.a.1 | $\langle -1, 13\rangle$ | 13 | 6 | II, $p = 5$ | I (pullback) |
| 43.132.9.a.2 | $\langle -1, 2\rangle$ | 9 | 7 | II+III, $p = 3$ | I (special) |
| 43.308.15.b.1 | $\langle -1, 6\rangle$ | 15 | 9 | II, $p = 2$ | I (special) |
| 44.144.8.c.4 | $\langle -1, 5\rangle$ | 8 | 5 | II, $p = 3$ | [37] |
| 44.360.16.e.1 | $\langle -1, 21\rangle$ | 16 | 8 | II, $p = 3$ | I (pullback) |
| 45.144.5.a.1 | $\langle -1, 4\rangle$ | 5 | 4 | II, $p = 2$ | [37] |
| 45.216.11.d.2 | $\langle -1, 8\rangle$ | 11 | 6 | II, $p = 2$ | I (pullback) |
| 45.288.9.a.4 | $\langle -1, 14\rangle$ | 9 | 6 | II, $p = 2$ | I (pullback) |
| 45.432.21.g.2 | $\langle -1, 19\rangle$ | 21 | 10 | II, $p = 2$ | I (cusps) |
| 48.192.5.ky.2 | $\langle -1, 11\rangle$ | 5 | 4 | II, $p = 5$ | I (pullback) |
| 48.192.7.hj.1 | $\langle -1, 5\rangle$ | 7 | 4 | II, $p = 5$ | I (pullback) |
| 48.192.7.ho.1 | $\langle -1, 7, 17\rangle$ | 7 | 4 | II, $p = 5$ | I (pullback) |
| 48.384.13.nt.1 | $\langle -1, 23\rangle$ | 13 | 8 | II, $p = 7$ | I (pullback) |
| 48.384.19.bc.1 | $\langle -1, 7\rangle$ | 19 | 8 | II, $p = 7$ | I (pullback) |
| 48.384.19.bj.1 | $\langle -1, 17\rangle$ | 19 | 8 | II, $p = 5$ | I (pullback) |
| 49.168.3.b.1 | $\langle -1, 6\rangle$ | 3 | 3 | genus | genus |
| 49.392.19.i.2 | $\langle -1, 18\rangle$ | 19 | 11 | II, $p = 2$ | I (cusps) |
| 50.180.4.a.2 | $\langle -1, 9\rangle$ | 4 | 3 | genus | I (pullback) |
| 50.450.22.f.1 | $\langle -1, 7\rangle$ | 22 | 10 | II, $p = 3$ | I (pullback) |

Table 4: Positive strict intermediate pairs.

| $N$ | Generators of $\Delta$ | $g$ | Proof |
|---|---|---|---|
| 13 | $\langle -1, 3\rangle$ | 0 | L |
| 13 | $\langle -1, 5\rangle$ | 0 | L |
| 15 | $\langle -1, 4\rangle$ | 1 | L |
| 16 | $\langle -1, 7\rangle$ | 0 | L |

| $N$ | Generators of $\Delta$ | $g$ | Proof |
|---|---|---|---|
| 17 | $\langle -1, 2\rangle$ | 1 | L |
| 17 | $\langle -1, 4\rangle$ | 1 | L |
| 19 | $\langle -1, 7\rangle$ | 1 | L |
| 20 | $\langle -1, 9\rangle$ | 1 | L |
| 21 | $\langle -1, 4\rangle$ | 1 | L |
| 21 | $\langle -1, 8\rangle$ | 3 | K |
| 24 | $\langle -1, 5\rangle$ | 3 | K |
| 24 | $\langle -1, 7\rangle$ | 3 | K |
| 24 | $\langle -1, 11\rangle$ | 1 | L |
| 25 | $\langle -1, 4\rangle$ | 0 | L |
| 25 | $\langle -1, 7\rangle$ | 4 | K |
| 26 | $\langle -1, 3\rangle$ | 4 | K |
| 26 | $\langle -1, 5\rangle$ | 4 | K |
| 27 | $\langle -1, 8\rangle$ | 1 | L |
| 28 | $\langle -1, 3\rangle$ | 4 | K |
| 28 | $\langle -1, 13\rangle$ | 4 | K |
| 29 | $\langle -1, 4\rangle$ | 4 | K |
| 30 | $\langle -1, 11\rangle$ | 5 | W |
| 31 | $\langle -1, 2\rangle$ | 6 | P |
| 31 | $\langle -1, 5\rangle$ | 6 | P |
| 32 | $\langle -1, 7\rangle$ | 1 | L |
| 32 | $\langle -1, 15\rangle$ | 5 | W |
| 33 | $\langle -1, 2\rangle$ | 5 | W |
| 34 | $\langle -1, 9\rangle$ | 5 | W |
| 35 | $\langle -1, 4\rangle$ | 5 | W |
| 36 | $\langle -1, 11\rangle$ | 3 | K |
| 37 | $\langle -1, 3\rangle$ | 4 | K |
| 37 | $\langle -1, 8\rangle$ | 4 | K |
| 39 | $\langle -1, 4\rangle$ | 5 | W |
| 40 | $\langle -1, 7\rangle$ | 7 | W |
| 40 | $\langle -1, 9, 11\rangle$ | 5 | W |
| 41 | $\langle -1, 2\rangle$ | 5 | W |
| 44 | $\langle -1, 5\rangle$ | 8 | P |
| 45 | $\langle -1, 4\rangle$ | 5 | W |
| 48 | $\langle -1, 11\rangle$ | 5 | W |
| 49 | $\langle -1, 6\rangle$ | 3 | K |
| 50 | $\langle -1, 9\rangle$ | 4 | K |
| 64 | $\langle -1, 7\rangle$ | 5 | W |
| 125 | $\langle -1, 4\rangle$ | 16 | P |

Table 5: The 154 strict pairs with a direct negative argument: 139 curves with an individual argument and the 15 curves of large index (code S, $119\mu > 120000$). Projections are listed separately.

| $N$ | Generators of $\Delta$ | $g$ | $\mu$ | Proof |
|---|---|---|---|---|
| 29 | $\langle -1, 12\rangle$ | 8 | 210 | Z |
| 33 | $\langle -1, 10\rangle$ | 11 | 240 | Z |
| 34 | $\langle -1, 13\rangle$ | 9 | 216 | Z |
| 35 | $\langle -1, 8\rangle$ | 7 | 144 | B |
| 35 | $\langle -1, 11\rangle$ | 9 | 192 | Z |

| $N$ | Generators of $\Delta$ | $g$ | $\mu$ | Proof |
|---|---|---|---|---|
| 35 | $\langle -1, 6\rangle$ | 13 | 288 | Z |
| 36 | $\langle -1, 17\rangle$ | 7 | 216 | T |
| 37 | $\langle -1, 10\rangle$ | 10 | 228 | D |
| 37 | $\langle -1, 6\rangle$ | 16 | 342 | E |
| 38 | $\langle -1, 7\rangle$ | 10 | 180 | Z |
| 39 | $\langle -1, 5\rangle$ | 9 | 168 | T |
| 39 | $\langle -1, 16\rangle$ | 9 | 224 | Z |
| 39 | $\langle -1, 14\rangle$ | 17 | 336 | Z |
| 40 | $\langle -1, 3\rangle$ | 7 | 144 | R |
| 40 | $\langle -1, 9\rangle$ | 13 | 288 | Z |
| 40 | $\langle -1, 11\rangle$ | 13 | 288 | Z |
| 40 | $\langle -1, 19\rangle$ | 9 | 288 | Z |
| 41 | $\langle -1, 4\rangle$ | 11 | 168 | Z |
| 41 | $\langle -1, 3\rangle$ | 11 | 210 | Z |
| 41 | $\langle -1, 9\rangle$ | 21 | 420 | Z |
| 42 | $\langle -1, 5\rangle$ | 9 | 192 | Z |
| 42 | $\langle -1, 13\rangle$ | 13 | 288 | Z |
| 43 | $\langle -1, 2\rangle$ | 9 | 132 | C |
| 43 | $\langle -1, 6\rangle$ | 15 | 308 | E |
| 44 | $\langle -1, 21\rangle$ | 16 | 360 | Z |
| 45 | $\langle -1, 8\rangle$ | 11 | 216 | Z |
| 45 | $\langle -1, 14\rangle$ | 9 | 288 | Z |
| 45 | $\langle -1, 19\rangle$ | 21 | 432 | Z |
| 48 | $\langle -1, 5\rangle$ | 7 | 192 | R |
| 48 | $\langle -1, 7, 17\rangle$ | 7 | 192 | R |
| 48 | $\langle -1, 7\rangle$ | 19 | 384 | Z |
| 48 | $\langle -1, 17\rangle$ | 19 | 384 | Z |
| 48 | $\langle -1, 23\rangle$ | 13 | 384 | Z |
| 49 | $\langle -1, 18\rangle$ | 19 | 392 | Z |
| 50 | $\langle -1, 7\rangle$ | 22 | 450 | Z |
| 51 | $\langle -1, 2\rangle$ | 9 | 144 | Z |
| 51 | $\langle -1, 4\rangle$ | 17 | 288 | Z |
| 51 | $\langle -1, 16\rangle$ | 33 | 576 | Z |
| 52 | $\langle -1, 3\rangle$ | 9 | 168 | Z |
| 52 | $\langle -1, 5\rangle$ | 13 | 252 | Z |
| 52 | $\langle -1, 9\rangle$ | 19 | 336 | Z |
| 52 | $\langle -1, 25\rangle$ | 25 | 504 | Z |
| 53 | $\langle -1, 4\rangle$ | 8 | 108 | F |
| 53 | $\langle -1, 23\rangle$ | 40 | 702 | E |
| 54 | $\langle -1, 17\rangle$ | 10 | 324 | Z |
| 55 | $\langle -1, 4\rangle$ | 9 | 144 | B |
| 55 | $\langle -1, 16\rangle$ | 17 | 288 | Z |
| 55 | $\langle -1, 12\rangle$ | 21 | 360 | Z |
| 55 | $\langle -1, 21\rangle$ | 41 | 720 | Z |
| 56 | $\langle -1, 3\rangle$ | 9 | 192 | Z |
| 56 | $\langle -1, 5\rangle$ | 11 | 192 | Z |
| 56 | $\langle -1, 17\rangle$ | 11 | 192 | Z |
| 56 | $\langle -1, 13, 15\rangle$ | 13 | 288 | Z |
| 56 | $\langle -1, 9\rangle$ | 21 | 384 | Z |
| 56 | $\langle -1, 13\rangle$ | 31 | 576 | Z |
| 56 | $\langle -1, 15\rangle$ | 31 | 576 | Z |

| $N$ | Generators of $\Delta$ | $g$ | $\mu$ | Proof |
|---|---|---|---|---|
| 56 | $\langle -1, 27\rangle$ | 25 | 576 | Z |
| 57 | $\langle -1, 2\rangle$ | 9 | 160 | D |
| 57 | $\langle -1, 11\rangle$ | 13 | 240 | E |
| 57 | $\langle -1, 7\rangle$ | 25 | 480 | E |
| 57 | $\langle -1, 20\rangle$ | 43 | 720 | E |
| 58 | $\langle -1, 5\rangle$ | 12 | 180 | E |
| 58 | $\langle -1, 17\rangle$ | 36 | 630 | E |
| 61 | $\langle -1, 4\rangle$ | 8 | 124 | F |
| 61 | $\langle -1, 8\rangle$ | 12 | 186 | E |
| 61 | $\langle -1, 21\rangle$ | 16 | 310 | E |
| 61 | $\langle -1, 3\rangle$ | 26 | 372 | E |
| 61 | $\langle -1, 13\rangle$ | 36 | 620 | E |
| 61 | $\langle -1, 11\rangle$ | 56 | 930 | E |
| 63 | $\langle -1, 4, 5\rangle$ | 9 | 192 | Z |
| 63 | $\langle -1, 2\rangle$ | 17 | 288 | Z |
| 63 | $\langle -1, 11\rangle$ | 17 | 288 | Z |
| 63 | $\langle -1, 10\rangle$ | 17 | 288 | Z |
| 63 | $\langle -1, 13\rangle$ | 13 | 288 | Z |
| 63 | $\langle -1, 4\rangle$ | 33 | 576 | Z |
| 63 | $\langle -1, 17\rangle$ | 33 | 576 | Z |
| 63 | $\langle -1, 20\rangle$ | 25 | 576 | Z |
| 63 | $\langle -1, 8\rangle$ | 49 | 864 | Z |
| 64 | $\langle -1, 15\rangle$ | 13 | 384 | Z |
| 64 | $\langle -1, 31\rangle$ | 37 | 768 | Z |
| 65 | $\langle -1, 2, 7\rangle$ | 9 | 168 | F |
| 65 | $\langle -1, 3\rangle$ | 11 | 168 | F |
| 65 | $\langle -1, 6\rangle$ | 11 | 168 | F |
| 65 | $\langle -1, 8, 12\rangle$ | 13 | 252 | E |
| 65 | $\langle -1, 2\rangle$ | 19 | 336 | E |
| 65 | $\langle -1, 7\rangle$ | 19 | 336 | E |
| 65 | $\langle -1, 9\rangle$ | 21 | 336 | E |
| 65 | $\langle -1, 8, 14\rangle$ | 25 | 504 | E |
| 65 | $\langle -1, 12\rangle$ | 31 | 504 | E |
| 65 | $\langle -1, 21\rangle$ | 31 | 504 | E |
| 65 | $\langle -1, 4\rangle$ | 41 | 672 | E |
| 65 | $\langle -1, 8\rangle$ | 55 | 1008 | E |
| 65 | $\langle -1, 14\rangle$ | 61 | 1008 | E |
| 65 | $\langle -1, 18\rangle$ | 55 | 1008 | E |
| 67 | $\langle -1, 3\rangle$ | 15 | 204 | E |
| 67 | $\langle -1, 29\rangle$ | 45 | 748 | E |
| 68 | $\langle -1, 9\rangle$ | 13 | 216 | Z |
| 72 | $\langle -1, 5\rangle$ | 13 | 288 | Z |
| 72 | $\langle -1, 7\rangle$ | 13 | 288 | Z |
| 72 | $\langle -1, 11\rangle$ | 9 | 288 | Z |
| 72 | $\langle -1, 17, 19\rangle$ | 21 | 432 | Z |
| 72 | $\langle -1, 23\rangle$ | 25 | 576 | Z |
| 72 | $\langle -1, 17\rangle$ | 49 | 864 | Z |
| 72 | $\langle -1, 19\rangle$ | 49 | 864 | Z |
| 72 | $\langle -1, 35\rangle$ | 41 | 864 | Z |
| 73 | $\langle -1, 6\rangle$ | 9 | 148 | D |
| 73 | $\langle -1, 7\rangle$ | 13 | 222 | E |

| $N$ | Generators of $\Delta$ | $g$ | $\mu$ | Proof |
|---|---|---|---|---|
| 73 | $\langle -1, 2\rangle$ | 19 | 296 | E |
| 73 | $\langle -1, 3\rangle$ | 25 | 444 | E |
| 73 | $\langle -1, 10\rangle$ | 43 | 666 | E |
| 73 | $\langle -1, 8\rangle$ | 55 | 888 | E |
| 73 | $\langle -1, 27\rangle$ | 85 | 1332 | S |
| 75 | $\langle -1, 4\rangle$ | 9 | 240 | T |
| 75 | $\langle -1, 14\rangle$ | 17 | 480 | Z |
| 75 | $\langle -1, 7\rangle$ | 37 | 600 | Z |
| 75 | $\langle -1, 26\rangle$ | 73 | 1200 | S |
| 80 | $\langle -1, 3, 13\rangle$ | 15 | 288 | Z |
| 80 | $\langle -1, 7, 9\rangle$ | 15 | 288 | Z |
| 80 | $\langle -1, 9, 11\rangle$ | 13 | 288 | Z |
| 80 | $\langle -1, 7\rangle$ | 35 | 576 | Z |
| 80 | $\langle -1, 13\rangle$ | 35 | 576 | Z |
| 80 | $\langle -1, 17\rangle$ | 35 | 576 | Z |
| 80 | $\langle -1, 9\rangle$ | 69 | 1152 | S |
| 80 | $\langle -1, 31\rangle$ | 69 | 1152 | S |
| 80 | $\langle -1, 39\rangle$ | 57 | 1152 | S |
| 81 | $\langle -1, 8\rangle$ | 10 | 324 | Z |
| 81 | $\langle -1, 26\rangle$ | 46 | 972 | Z |
| 91 | $\langle -1, 3, 4\rangle$ | 13 | 224 | E |
| 91 | $\langle -1, 2\rangle$ | 21 | 336 | E |
| 91 | $\langle -1, 11\rangle$ | 21 | 336 | E |
| 91 | $\langle -1, 5\rangle$ | 23 | 336 | E |
| 91 | $\langle -1, 6\rangle$ | 23 | 336 | E |
| 91 | $\langle -1, 9, 12\rangle$ | 25 | 448 | E |
| 91 | $\langle -1, 3\rangle$ | 41 | 672 | E |
| 91 | $\langle -1, 4\rangle$ | 41 | 672 | E |
| 91 | $\langle -1, 25\rangle$ | 45 | 672 | E |
| 91 | $\langle -1, 36\rangle$ | 45 | 672 | E |
| 91 | $\langle -1, 8\rangle$ | 67 | 1008 | E |
| 91 | $\langle -1, 9\rangle$ | 81 | 1344 | S |
| 91 | $\langle -1, 12\rangle$ | 89 | 1344 | S |
| 91 | $\langle -1, 16\rangle$ | 81 | 1344 | S |
| 91 | $\langle -1, 22\rangle$ | 89 | 1344 | S |
| 91 | $\langle -1, 27\rangle$ | 133 | 2016 | S |
| 109 | $\langle -1, 3\rangle$ | 16 | 220 | E0 |
| 109 | $\langle -1, 2\rangle$ | 22 | 330 | E0 |
| 109 | $\langle -1, 4\rangle$ | 46 | 660 | E0 |
| 109 | $\langle -1, 8\rangle$ | 64 | 990 | E0 |
| 109 | $\langle -1, 45\rangle$ | 136 | 1980 | S |
| 109 | $\langle -1, 33\rangle$ | 208 | 2970 | S |
| 121 | $\langle -1, 10\rangle$ | 26 | 660 | E |
| 121 | $\langle -1, 3\rangle$ | 106 | 1452 | S |
| 125 | $\langle -1, 7\rangle$ | 36 | 750 | E0 |
| 125 | $\langle -1, 24\rangle$ | 76 | 1500 | S |
| 125 | $\langle -1, 57\rangle$ | 256 | 3750 | S |

TABLE 6. Projections to intermediate curves (other than $X_0$) used in the proof of Theorem B.

| $N$ | Source generators | $M$ | Target generators |
|---|---|---|---|
| 63 | $\langle -1, 5\rangle$ | 63 | $\langle -1, 4, 5\rangle$ |
| 68 | $\langle -1, 13\rangle$ | 34 | $\langle -1, 13\rangle$ |
| 68 | $\langle -1, 33\rangle$ | 34 | $\langle -1, 13\rangle$ |
| 80 | $\langle -1, 3\rangle$ | 40 | $\langle -1, 3\rangle$ |
| 80 | $\langle -1, 9, 31\rangle$ | 40 | $\langle -1, 3\rangle$ |
| 80 | $\langle -1, 11\rangle$ | 40 | $\langle -1, 11\rangle$ |
| 80 | $\langle -1, 19\rangle$ | 40 | $\langle -1, 19\rangle$ |

TABLE 7. The twelve exceptional negative certificates.

| $N$ | Generators of $\Delta$ | $g$ | Method |
|---|---|---|---|
| 36 | $\langle -1, 17\rangle$ | 7 | Petri lifting |
| 39 | $\langle -1, 5\rangle$ | 9 | Petri lifting |
| 40 | $\langle -1, 3\rangle$ | 7 | Rational classes |
| 43 | $\langle -1, 2\rangle$ | 9 | Empty cosets |
| 48 | $\langle -1, 5\rangle$ | 7 | Rational classes |
| 48 | $\langle -1, 7, 17\rangle$ | 7 | Rational classes |
| 53 | $\langle -1, 4\rangle$ | 8 | Elliptic families |
| 61 | $\langle -1, 4\rangle$ | 8 | Elliptic families |
| 65 | $\langle -1, 2, 7\rangle$ | 9 | Elliptic families |
| 65 | $\langle -1, 3\rangle$ | 11 | Elliptic families |
| 65 | $\langle -1, 6\rangle$ | 11 | Elliptic families |
| 75 | $\langle -1, 4\rangle$ | 9 | Petri lifting |

DEPARTMENT OF MATHEMATICS, FACULTY OF SCIENCE, UNIVERSITY OF ZAGREB, BIJENIČKA CESTA 30, 10000 ZAGREB, CROATIA

*Email address*: marin.varivoda@math.hr

TABLE 8. Theorem A, Route T: the known family $V = B + [F_4] + C$, the subgroups $S$ tested over $\mathbb{F}_p$, and the counts. "Divisors" and "classes" are the effective divisors of degree 5 over $\mathbb{F}_p$ and their classes.

| $N$ | $p$ | quotient by | $\dim B$ | $\#B(\mathbb{F}_p)$ | subgroups tested | divisors | classes | cosets | $\dim TZ_S$ |
|---|---|---|---|---|---|---|---|---|---|
| 74 | 3 | $w_{74}$ | 2 | 21 | all of order 7 | 1340 | 1316 | 18 | 3 |
| 77 | 5 | $w_{77}$ | 2 | 35 | all of order 7 | 12528 | 12488 | 40 | 3 |
| 82 | 3 | $w_2, w_{41}$ | 1 | 6 | $B(\mathbb{F}_3)$ | 2574 | 2568 | 6 | 2 |
| 86 | 7 | $w_2, w_{43}$ | 1 | 8 | cyclic of order 8 (192) | 64384 | 64384 | 12 | 2 |
| 99 | 5 | $w_9, w_{11}$ | 1 | 10 | $B(\mathbb{F}_5)$ | 8504 | 8504 | 6 | 2 |
| 111 | 2 | $w_{111}$ | 2 | 15 | all of order 5 | 328 | 320 | 12 | 3 |
| 118 | 3 | $w_2, w_{59}$ | 1 | 5 | all of order 5 | 3104 | 3104 | 6 | 2 |
| 123 | 2 | $w_3, w_{41}$ | 1 | 3 | order 3 or 5 | 842 | 842 | 6 | 2 |
| 128 | 3 | $w_{128}, sw_{128}s^{-1}$ | 1 | 6 | $B(\mathbb{F}_3)$ | 1812 | 1812 | 4 | 2 |
| 141 | 2 | $w_3, w_{47}$ | 1 | 3 | order 3 or 5 | 566 | 566 | 4 | 2 |
| 142 | 3 | $w_{71}$ | 2 | 15 | order 5 or 7 | 1668 | 1656 | 12 | 3 |
| 143 | 2 | $w_{143}$ | 2 | 15 | all of order 3 | 332 | 324 | 20 | 3 |
| 145 | 3 | $w_5, w_{29}$ | 1 | 4 | all of order 2 | 1552 | 1552 | 24 | 2 |
| 155 | 2 | $w_5, w_{31}$ | 1 | 3 | order 3 or 5 | 536 | 536 | 4 | 2 |
| 159 | 5 | $w_3, w_{53}$ | 1 | 6 | all of order 3 | 23476 | 23476 | 16 | 2 |

TABLE 9. Theorem A, Route Q: $Y = X_0(N)/w_N$, $A = \pi^* \operatorname{Jac} Y$, $R = J/A$ with $R(\mathbb{Q})$ cyclic of order $m$, and the surviving indices $i$ of the translates $T_i = [5\infty] + it + A$.

| $N$ | $g$ | $g(Y)$ | $p$ | $m$ | $\#A(\mathbb{F}_p)$ | surviving $i$ |
|---|---|---|---|---|---|---|
| 79 | 6 | 1 | 3 | 13 | 5 | 1, 2, 3, 4 |
| 83 | 7 | 1 | 7 | 41 | 11 | 1, 2, 3, 4, 23 |
| 89 | 7 | 1 | 3 | 22 | 5 | 1, 2, 3, 4 |
| 101 | 8 | 1 | 5 | 25 | 7 | 1, 2, 3, 4 |
| 103 | 8 | 2 | 5 | 17 | 55 | 2, 3 |
| 107 | 9 | 2 | 3 | 53 | 29 | 2, 3 |
| 131 | 11 | 1 | 3 | 65 | 5 | 1, 2, 3, 4 |
| 167 | 14 | 2 | 3 | 83 | 19 | 2, 3 |
| 191 | 16 | 2 | 2 | 95 | 11 | 2, 3 |

TABLE 10. Theorem B, class E: for each level $N$ and index $t = [(\mathbb{Z}/N\mathbb{Z})^\times : \Delta]$, the number of curves, their genera, the elliptic curves of conductor $N$ with odd analytic rank, and their optimal modular degree $d_E$ (Cremona's database). At $N = 67, 73$ there are none, so these curves have no positive-rank elliptic quotient. Every positive-rank elliptic quotient has degree divisible by $t\,d_E$ [13, Corollary 4.10], which is composite, hence $\neq 5$.

| $N$ | $t$ | curves | genera | $E$ | $d_E$ |
|---|---|---|---|---|---|
| 37 | 9 | 1 | 16 | 37a1 | 2 |
| 43 | 7 | 1 | 15 | 43a1 | 2 |
| 53 | 13 | 1 | 40 | 53a1 | 2 |
| 57 | 3 | 1 | 13 | 57a1 | 4 |
| 57 | 6 | 1 | 25 | 57a1 | 4 |
| 57 | 9 | 1 | 43 | 57a1 | 4 |
| 58 | 2 | 1 | 12 | 58a1 | 4 |
| 58 | 7 | 1 | 36 | 58a1 | 4 |
| 61 | 3 | 1 | 12 | 61a1 | 2 |
| 61 | 5 | 1 | 16 | 61a1 | 2 |
| 61 | 6 | 1 | 26 | 61a1 | 2 |
| 61 | 10 | 1 | 36 | 61a1 | 2 |
| 61 | 15 | 1 | 56 | 61a1 | 2 |
| 65 | 3 | 1 | 13 | 65a1 | 2 |
| 65 | 4 | 3 | 19, 19, 21 | 65a1 | 2 |
| 65 | 6 | 3 | 25, 31, 31 | 65a1 | 2 |
| 65 | 8 | 1 | 41 | 65a1 | 2 |
| 65 | 12 | 3 | 55, 55, 61 | 65a1 | 2 |
| 67 | 3 | 1 | 15 | – | – |
| 67 | 11 | 1 | 45 | – | – |
| 73 | 3 | 1 | 13 | – | – |
| 73 | 4 | 1 | 19 | – | – |
| 73 | 6 | 1 | 25 | – | – |
| 73 | 9 | 1 | 43 | – | – |
| 73 | 12 | 1 | 55 | – | – |
| 91 | 2 | 1 | 13 | 91a1, 91b1 | 4 |
| 91 | 3 | 4 | 21, 21, 23, 23 | 91a1, 91b1 | 4 |
| 91 | 4 | 1 | 25 | 91a1, 91b1 | 4 |
| 91 | 6 | 4 | 41, 41, 45, 45 | 91a1, 91b1 | 4 |
| 91 | 9 | 1 | 67 | 91a1, 91b1 | 4 |
| 121 | 5 | 1 | 26 | 121b1 | 4 |